

Name:	ID:	Section:
-------	-----	----------

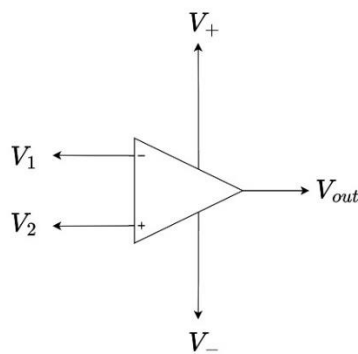
Problem Set 2 (Operational Amplifiers)

BRAC University

Semester: Summer 2024

Course No.: CSE251	Marks: 100
Course title: Electronic Devices and Circuits	Upload Date: 8/6/2024
Faculty: TMY	Submission Date: 6/7/2024

Operational Amplifier:



The operational amplifier is a device that takes two voltages as input and outputs the amplified difference of those two voltages.

The two inputs, V_1 (called the **inverting input**) and V_2 (called the **non-inverting input**) are subtracted to give

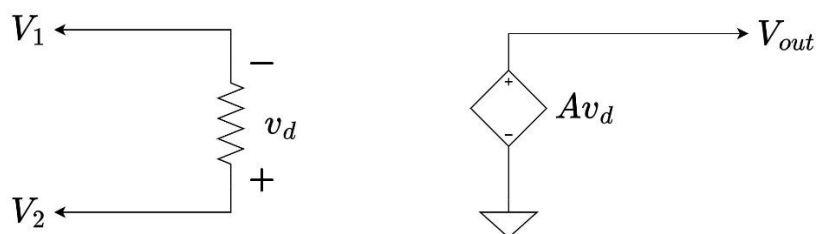
$$v_d = V_2 - V_1.$$

This difference voltage, v_d is then amplified to give the output,

$$V_{out} = A v_d = A(v_2 - v_1),$$

where A usually is a very large value ($A \sim 2 \times 10^5$).

An equivalent circuit considering only the input and output relations can be redrawn as:



However, the opamp cannot output beyond its connected power supplies, V_+ and V_- . Usually $V_+ > V_-$ and we avoid the inversion case, unless we really want to fry our IC!

Thus, our final mathematical representation of op amp's input output relation are as follows:

$$V_{out} = A(V_2 - V_1) \text{ where, } V_- < V_{out} < V_+$$

- 1) (CO1, 0.5 x 12 = 6) Geared with this idea, fillup the following table based on the given values of V_1 and V_2 . Assume $V_+ = +V_{sat} = +15V$ and $V_- = -V_{sat} = -15V$ where $V_{sat} = 15$ and $A = 2 \times 10^5$. The first are done for you:

V_1 (V)	V_2 (V)	V_d (V)	Av_d (V)	V_{out} (V)
10 μ V	15 μ V	5 μ V	1 V	1 V
10 mV	5 mV	-5 mV	-1000 V	-15 V
1 μ V	3 μ V			
0	5 μ V			
-10 μ V	-20 μ V			
1 mV	5 mV			
10 mV	2 mV			
1 V	2 V			
2 V	1 V			
1.001 V	1 V			
1 V	1.001 V			
2 mV	10 mV			
10 mV	2 mV			
10 mV	10 mV			

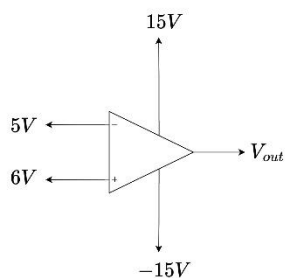
An interesting conclusion can be made from the table above. We see that for even the slightest difference in inputs, the output is either the maximum possible output voltage, +15V, or the minimum possible output voltage, -15V, depending on which input is bigger. Considering this, since we usually do not deal with such small voltages, we can approximate the operational amplifier output as:

$$\text{If } V_2 > V_1 \text{ then, } V_{out} = V_+$$

$$\text{If } V_2 < V_1 \text{ then, } V_{out} = V_-$$

And the output is capped between V_- and V_+ , i.e, $V_- < V_{out} < V_+$.

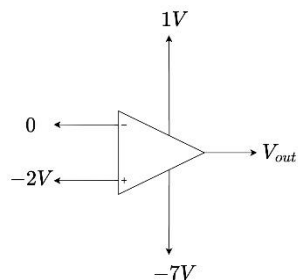
As an example, consider the circuit below:



Here, $V_2 = 6V$ and $V_1 = 5V$.

Since $V_2 > V_1$,

then the output, $V_{out} = V_+ = 15V$.



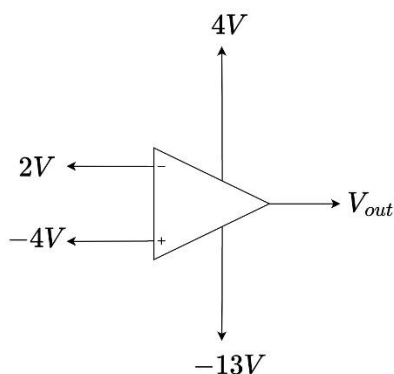
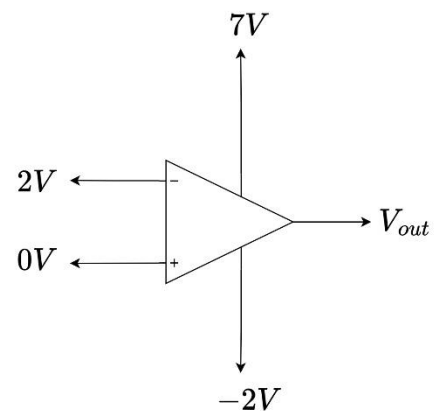
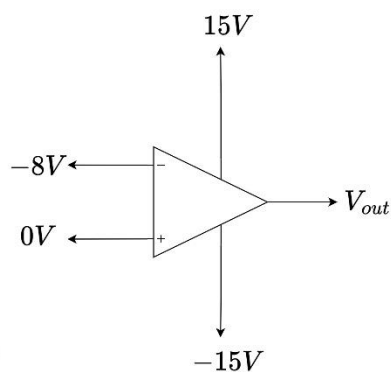
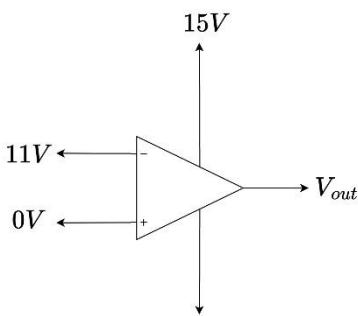
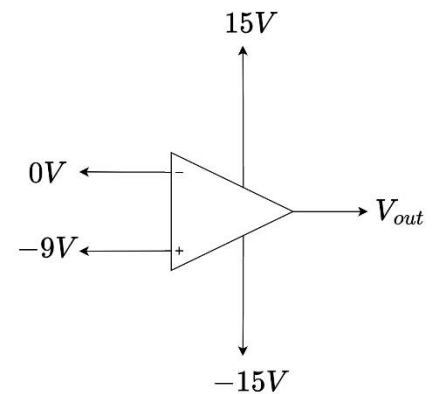
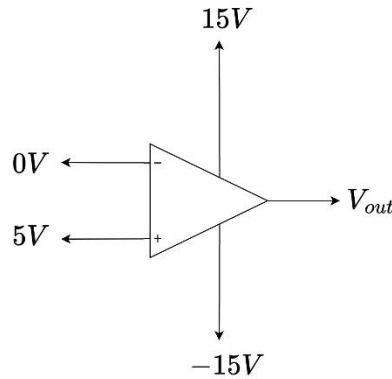
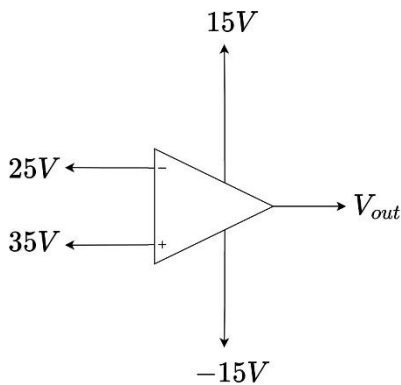
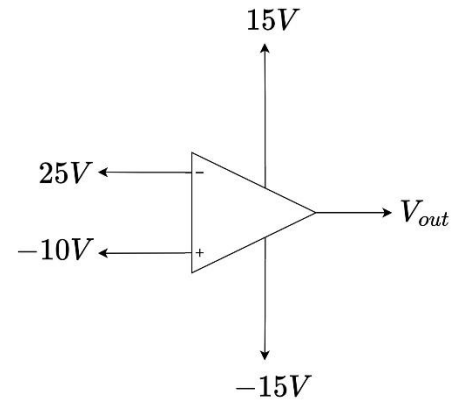
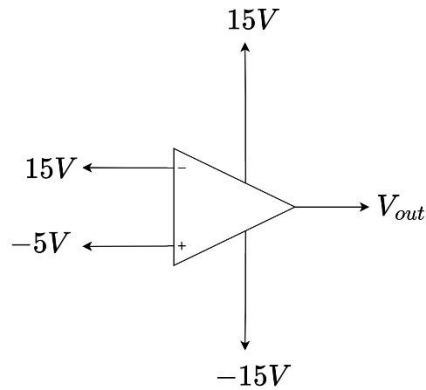
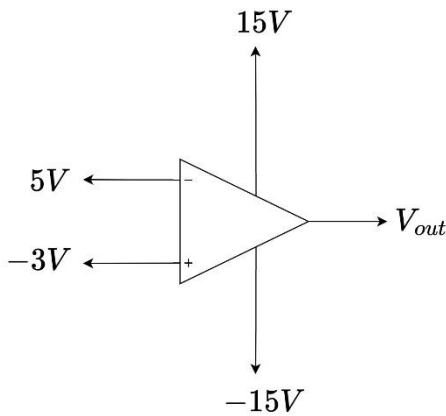
Here, $V_2 = -2V$ and $V_1 = 0V$.

Since $V_2 < V_1$,

then the output, $V_{out} = V_- = -7V$.

These circuits where two inputs are compared and then an output is obtained that is either the max value or the min value, is called the comparator circuit. The name stems from the idea that it compares two inputs and gives an output accordingly.

2) (CO1, 0.5 x 11 = 5.5) Find the output of the following comparator circuits with these ideas:



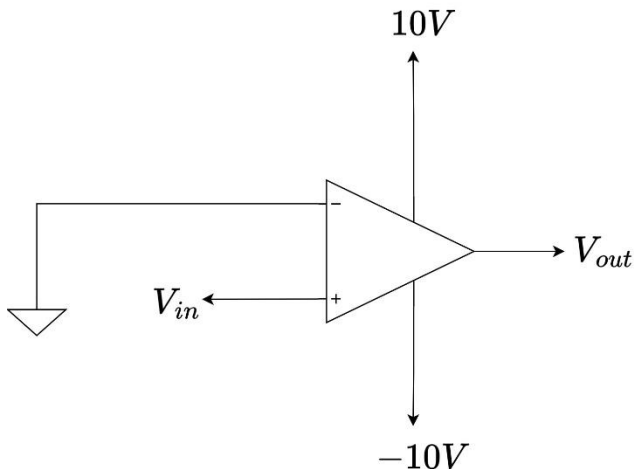
Thus, by simply checking if one input is bigger than the other, the output of the comparator can be found out as per its maximum and minimum values (**also called the saturation values**).

Max voltage of the op-amp as per the V_+ power supply: **Positive Saturation Voltage**

Min. voltage of the op-amp as per the V_- power supply: **Negative Saturation Voltage**

Now, let us fix one input voltage and vary another input with respect to time and see how we can draw the output waveform. Now, instead of considering instantaneous DC values, we will consider input and output waveforms. We will look at 3 cases and leave the fourth case for the reader to deduce on their own. For now, we will consider $V_{sat} = 10V$ and $V_+ = +V_{sat}$ and $V_- = -V_{sat}$ in all our cases for ease of graphing.

Case 1: $V_1 = 0$ (Ground the inverting input):



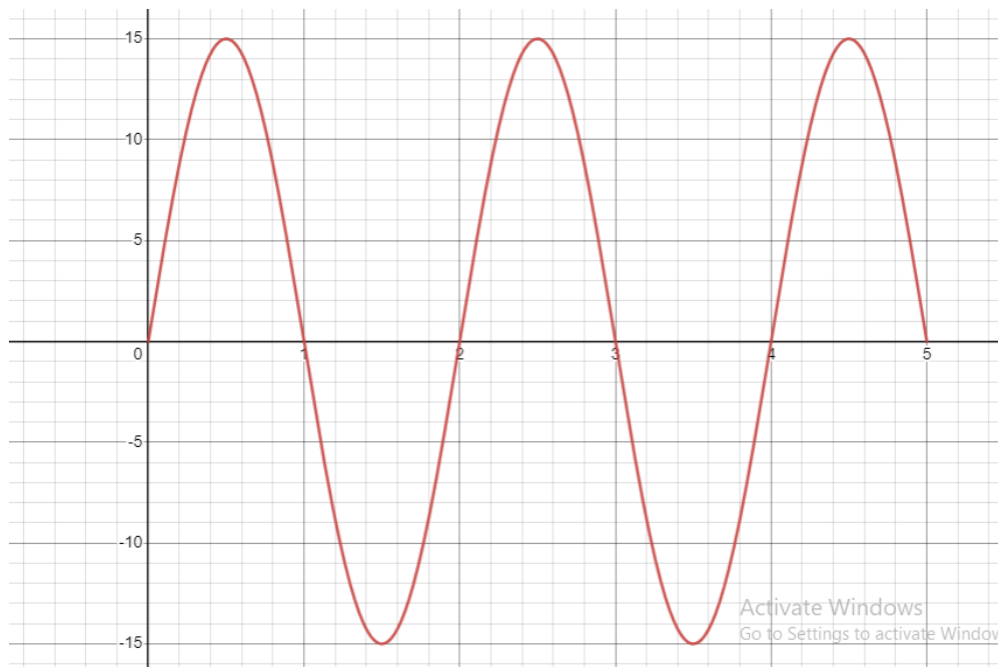
Here, $V_1 = 0V$ and $V_2 = V_{in}$. V_{in} will be a given input waveform that will change its values as time progresses. If we write down the mathematical expressions for our comparator circuit, we get:

$$V_2 > V_1: V_{out} = 10V \text{ and } V_2 < V_1: V_{out} = -10V$$

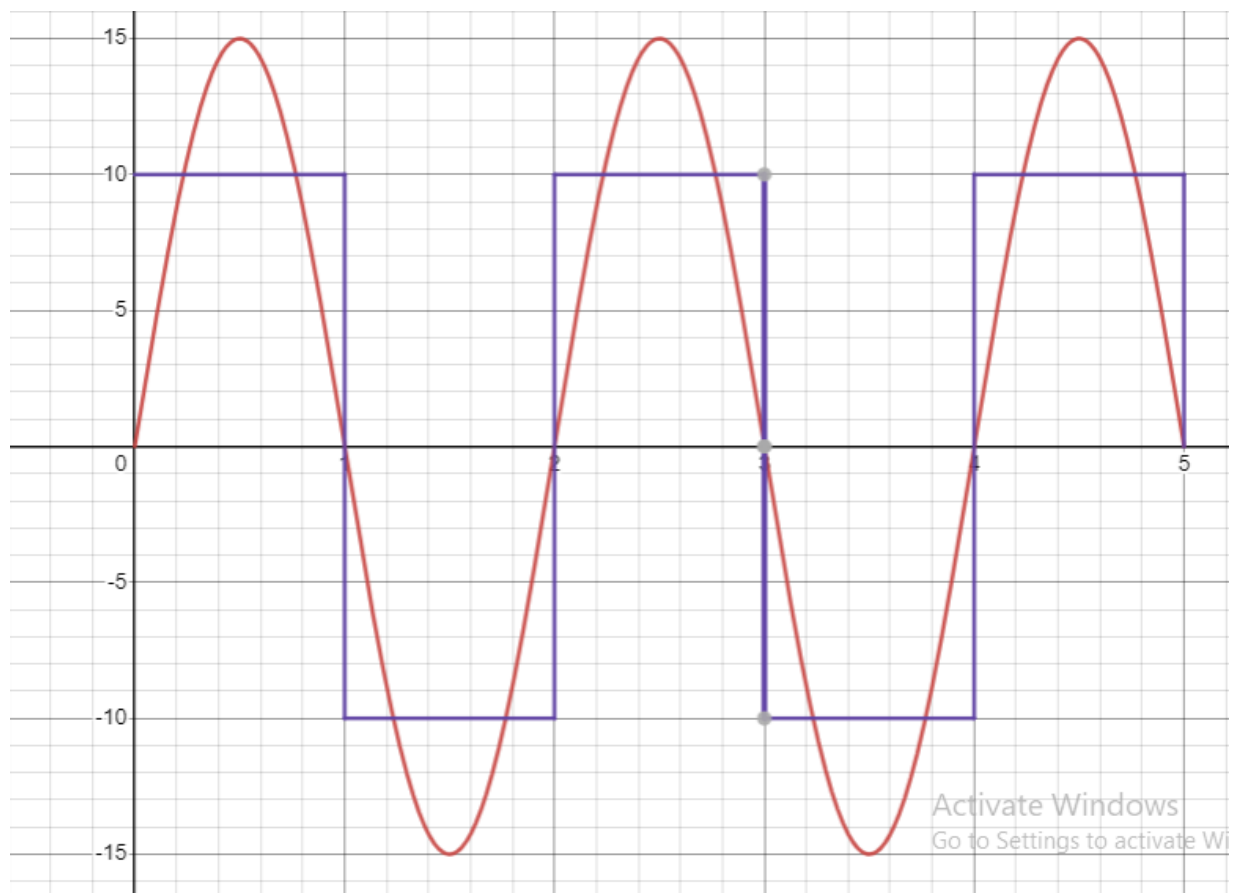
From our circuit, we can now write:

$$V_{in} > 0V: V_{out} = 10V \text{ and } V_{in} < 0V: V_{out} = -10V$$

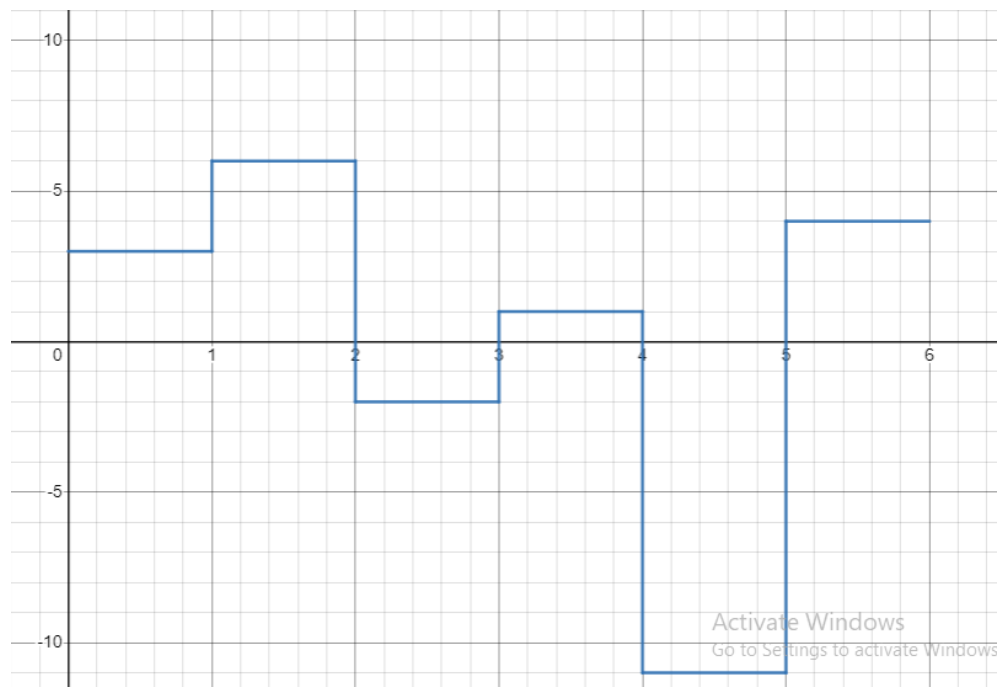
In other words, our output will always be $10V$ whenever input is positive (that is, greater than $0V$) and it will be $-10V$ whenever input is negative (that is, less than $0V$). Let us now take an input voltage that is an AC waveform:



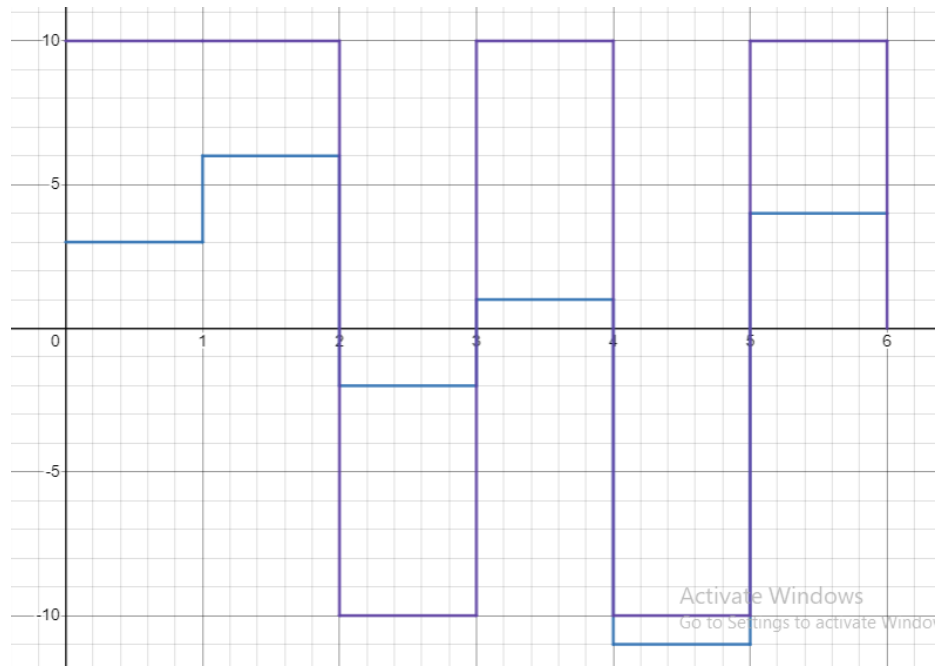
Say, the waveform on the left is our given input. What will be the output? A good way to answer this question is to first notice when this waveform is greater than zero, and when it's less than zero. Whenever it's greater than zero, output will be $10V$, and whenever it's less than zero, output will be $-10V$. This means that, initially, for the first one second, output will be $10V$. The next second, from $1s$ to $2s$, output will be $-10V$. Thus, the output can be drawn as:



Where the sinusoid is our input, V_{in} and the purple waveform is our output voltage, V_{out} . Here is a more tricky example:
Consider the same comparator circuit but with V_{in} as the input waveform given below:

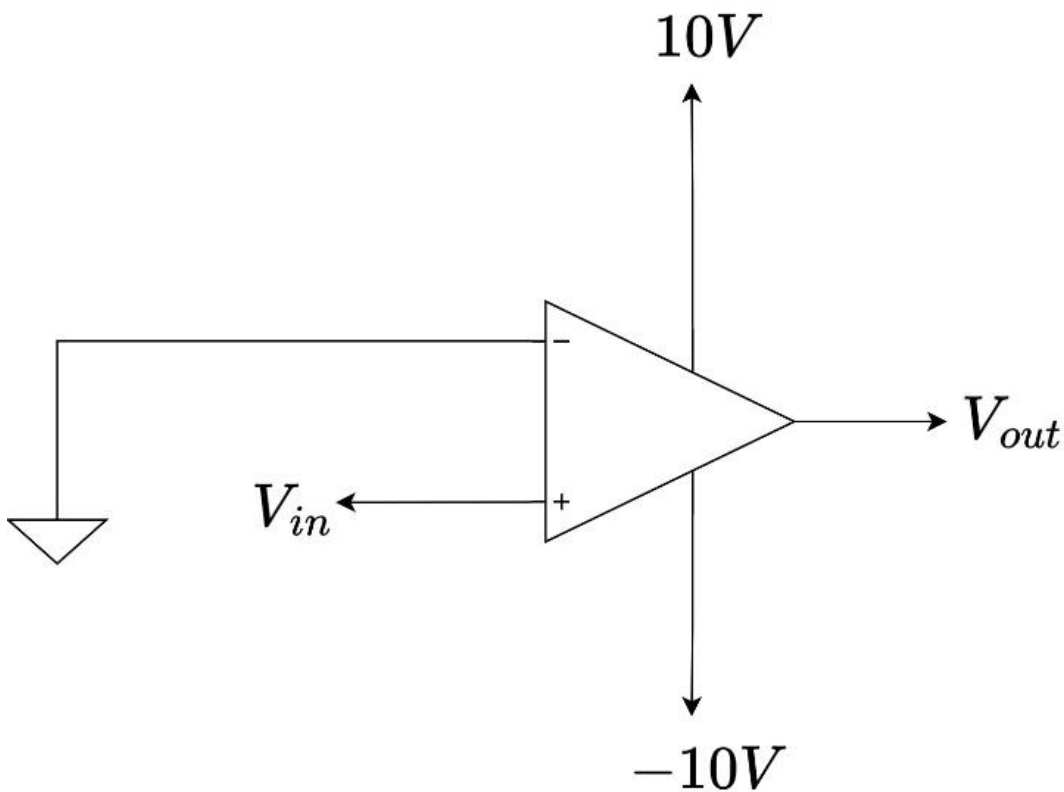


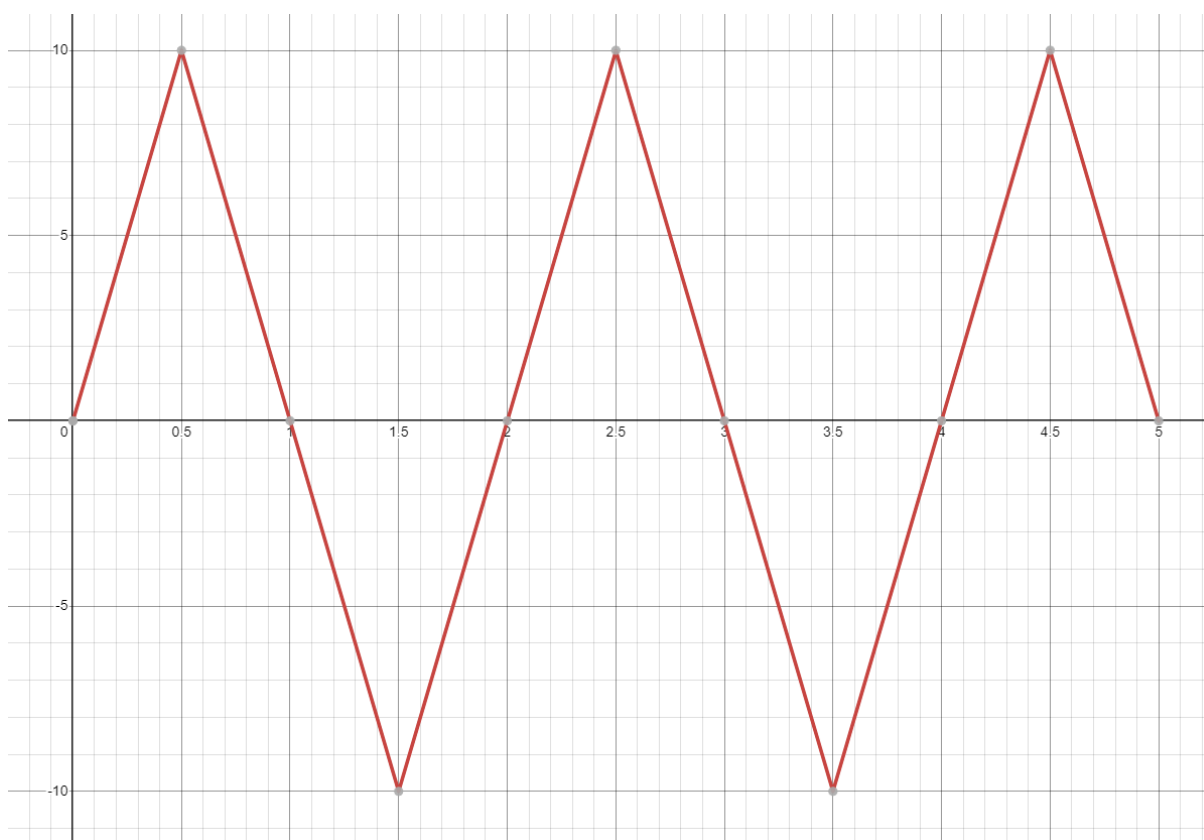
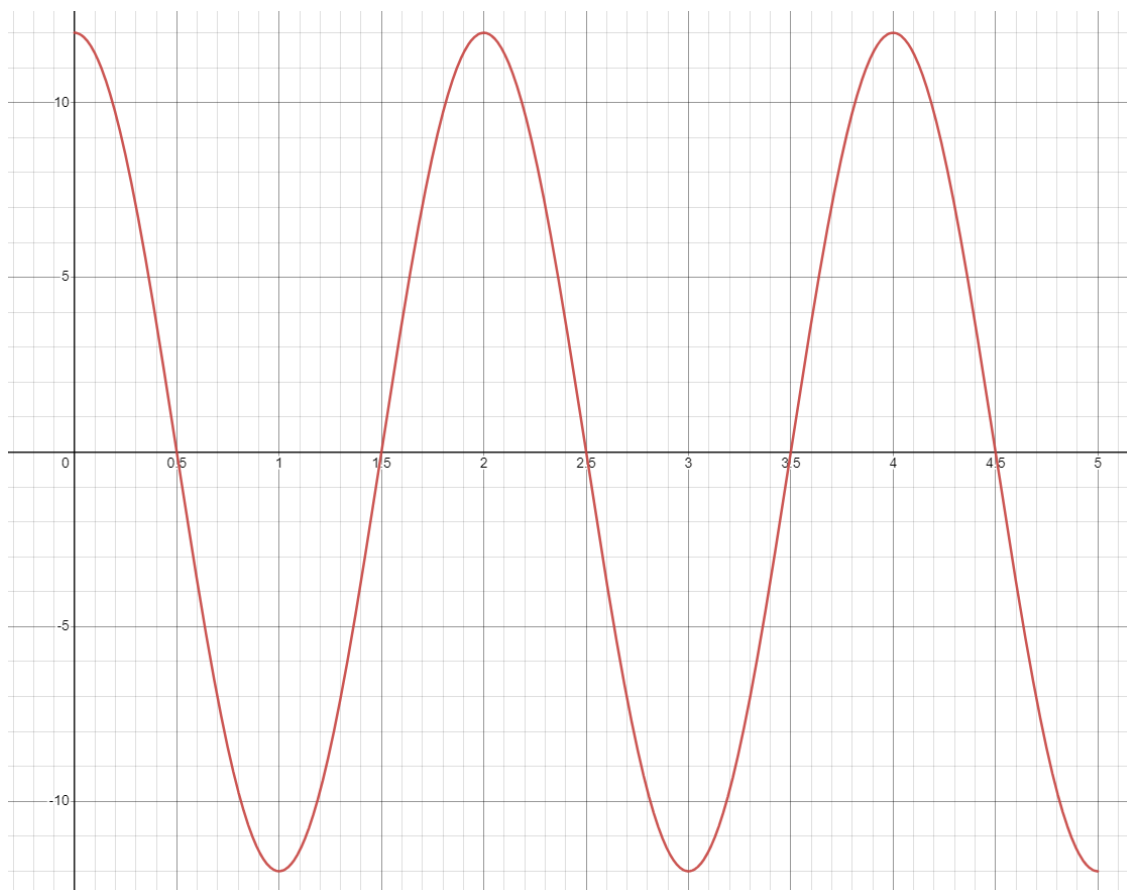
The waveform might look complicated, but all we have to do is check when its bigger or smaller than 0V and draw our output as per our comparator circuit relations. If we do so, we get our output as:

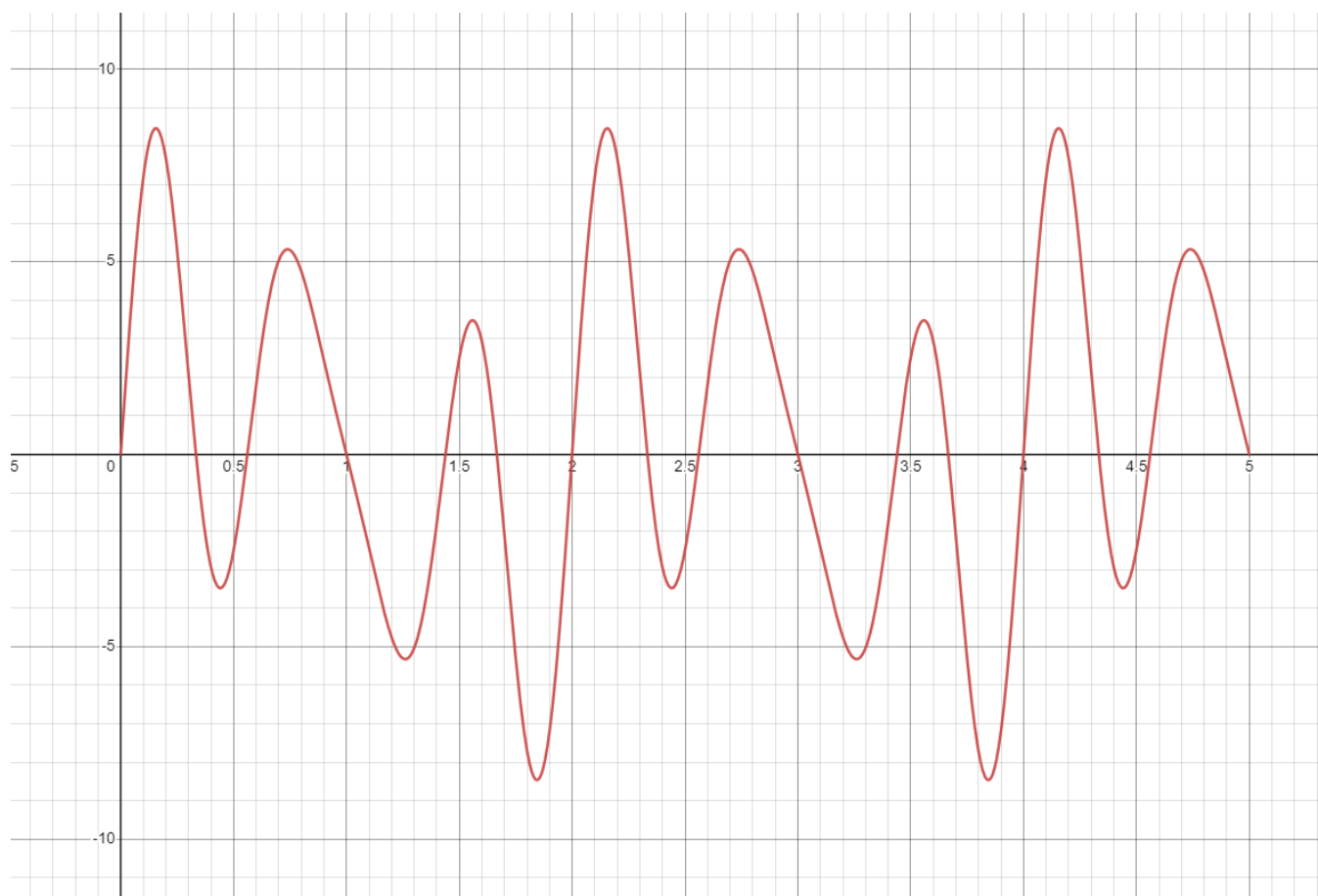
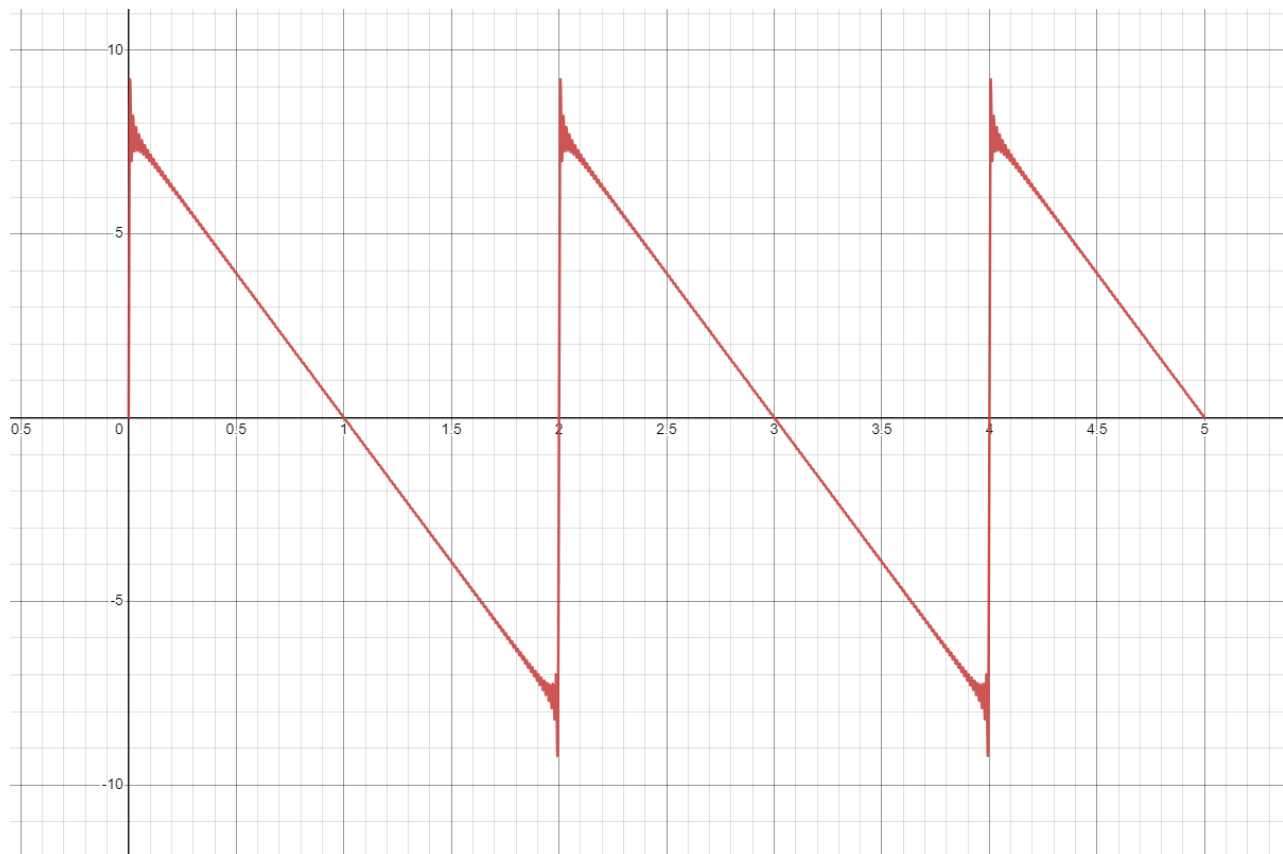


Regardless of what value the input voltage is at any time, if its bigger than 0V, output becomes 10V. If its smaller than 0V, output becomes -10V. Very simple, yet tricky at times!

- 3) (CO1, 2.5 x 4 = 10) Now find the output waveform (V_{out}) of the following input waveforms (V_{in}) as per the case 1 circuit we used (draw the output waveform on the graph in the question directly either by pen or pencil):

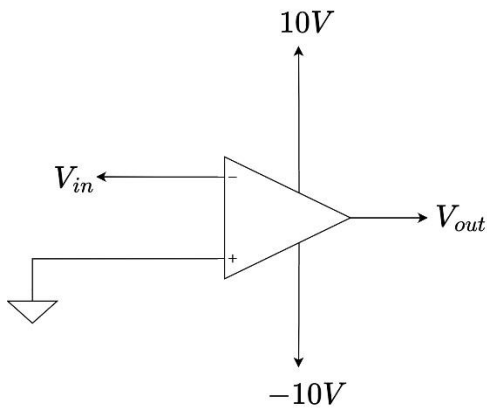






Case 2: $V_2 = 0$ (Grounding the noninverting input):

In this case, the roles of V_1 and V_2 are reversed. The circuit will now look like this:



Here, $V_1 = V_{in}$ and $V_2 = 0$. V_{in} will be a given input waveform that will change its values as time progresses. If we write down the mathematical expressions for our comparator circuit, we get:

$$V_2 > V_1: V_{out} = 10V \text{ and } V_2 < V_1: V_{out} = -10V$$

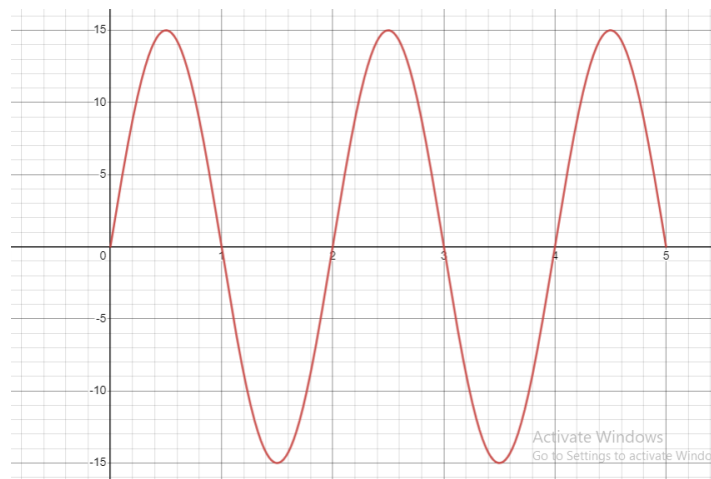
From our circuit, we can now write:

$$0V > V_{in}: V_{out} = 10V \text{ and } 0V < V_{in}: V_{out} = -10V$$

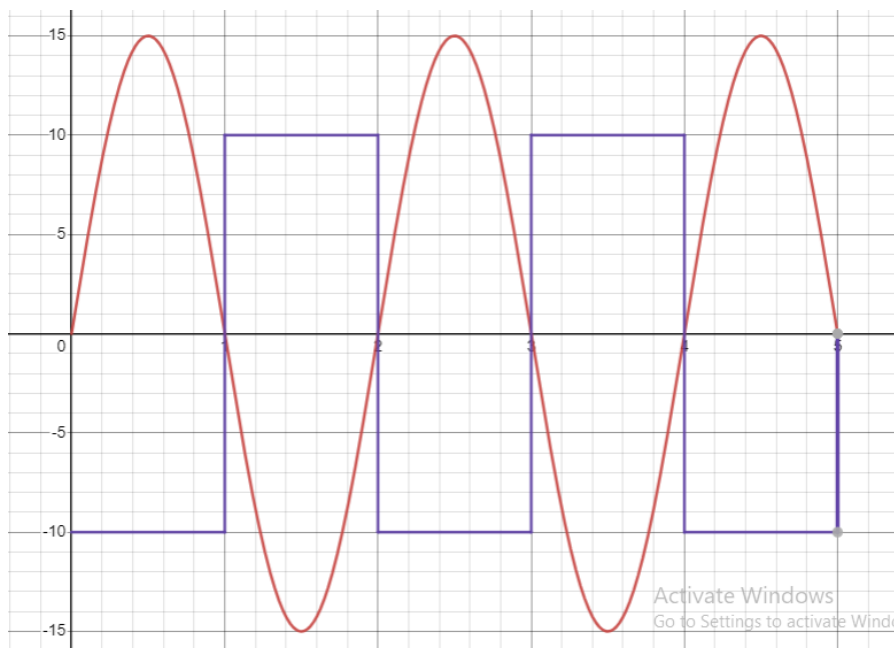
Which can be further rewritten as:

$$V_{in} < 0V: V_{out} = 10V \text{ and } V_{in} > 0V: V_{out} = -10V$$

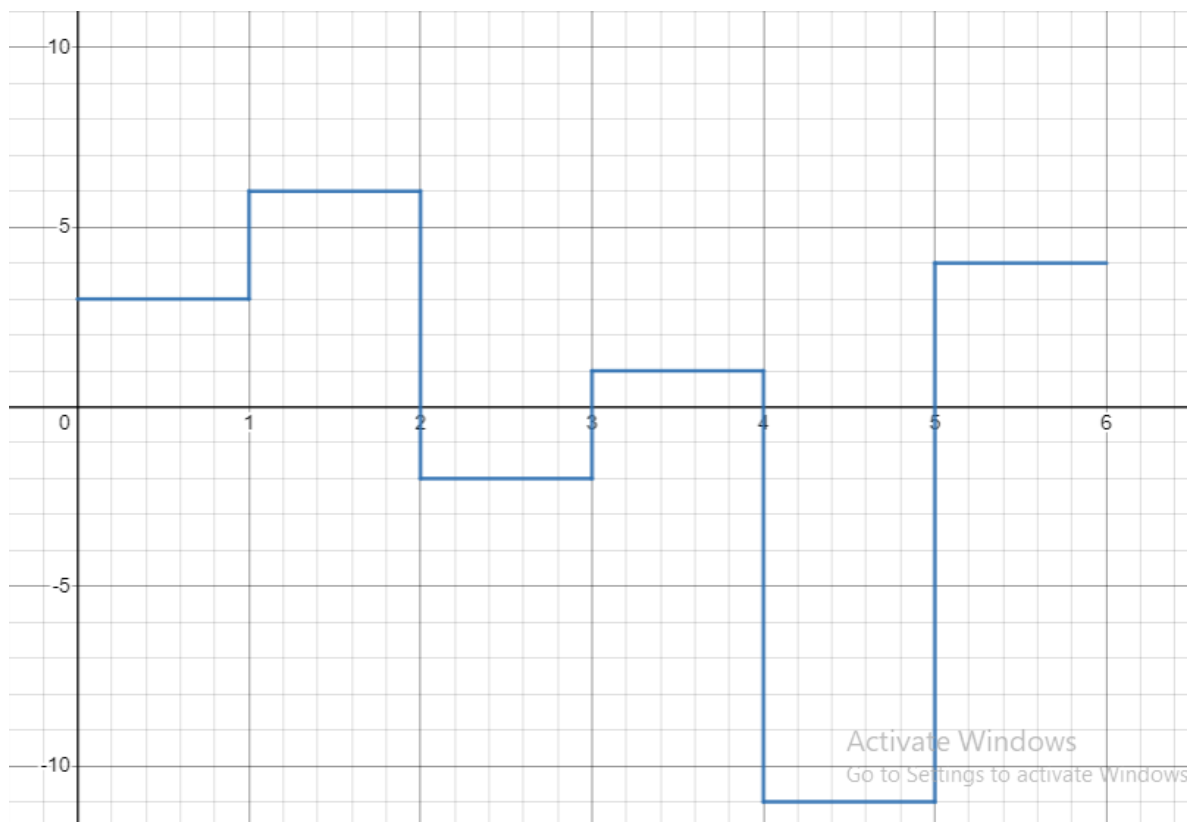
In other words, our output will always be $10V$ whenever input is negative (that is, less than $0V$) and it will be $-10V$ whenever input is positive (that is, greater than $0V$). This is the exact opposite of case 1. Let us now take an input voltage that is an AC waveform:



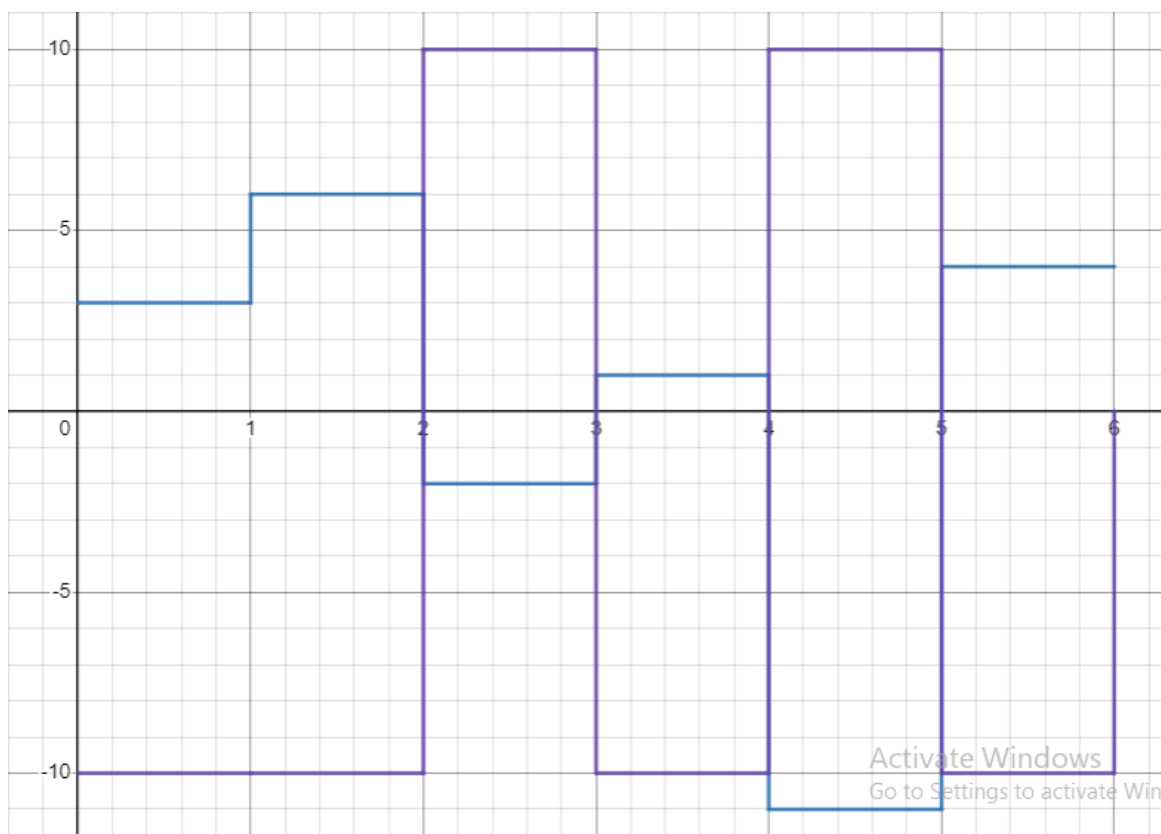
Now the negative portions will cause output to go to max value and the positive portions will cause output to go to min value thus giving us the following output graph:



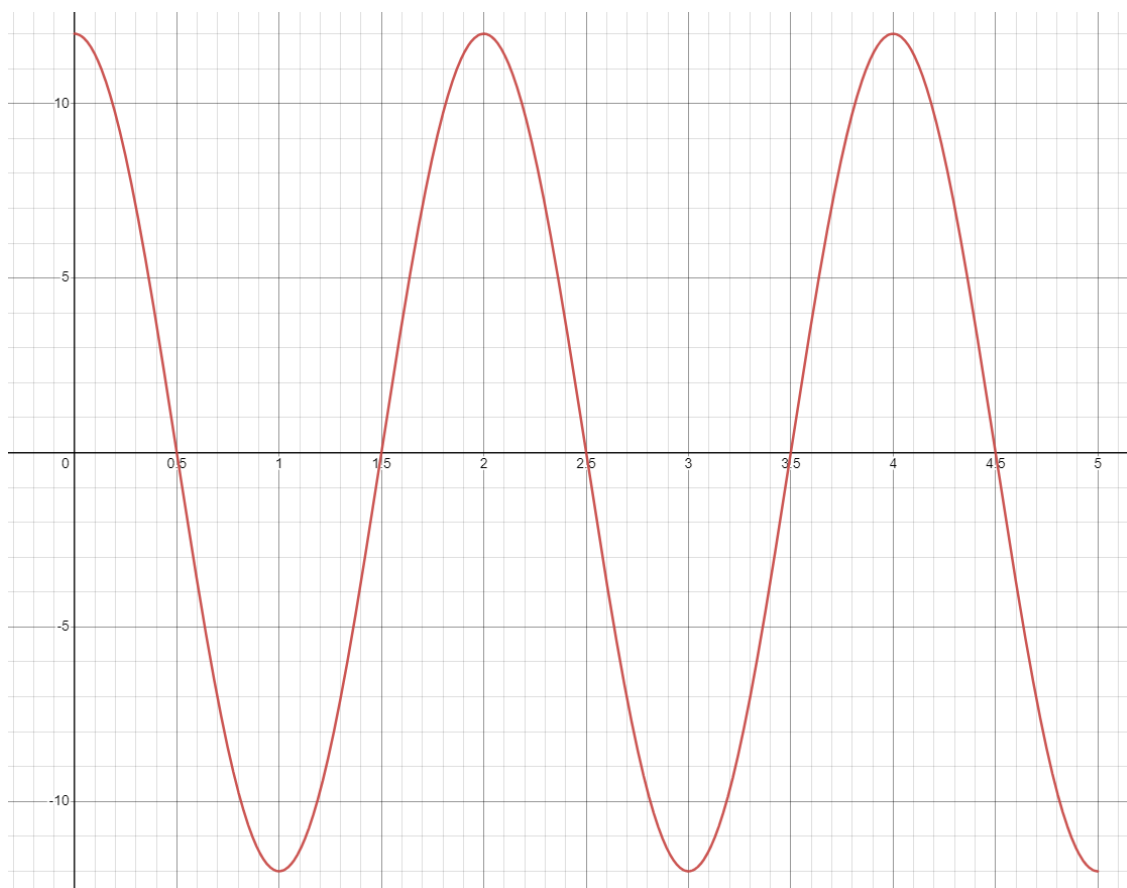
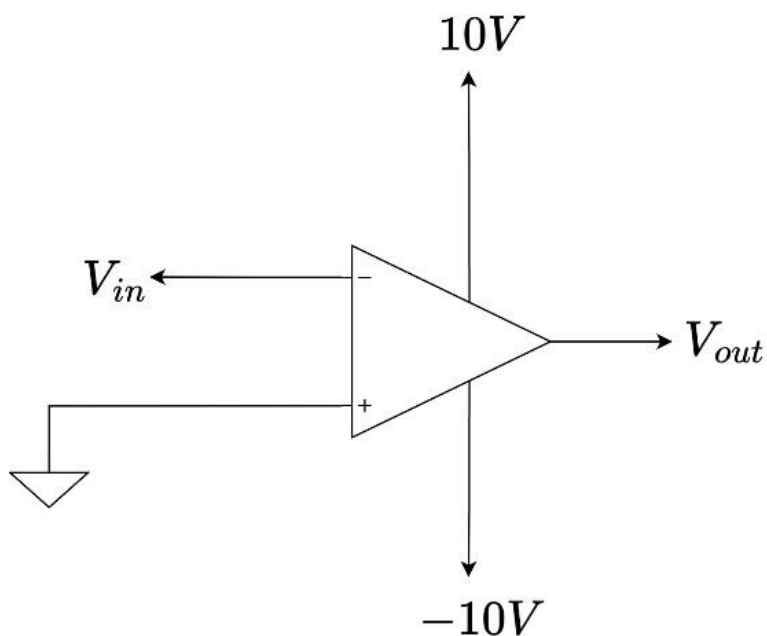
Where the purple waveform is the output voltage and the red waveform is our input voltage. Let us also see the output for our previously tricky input:

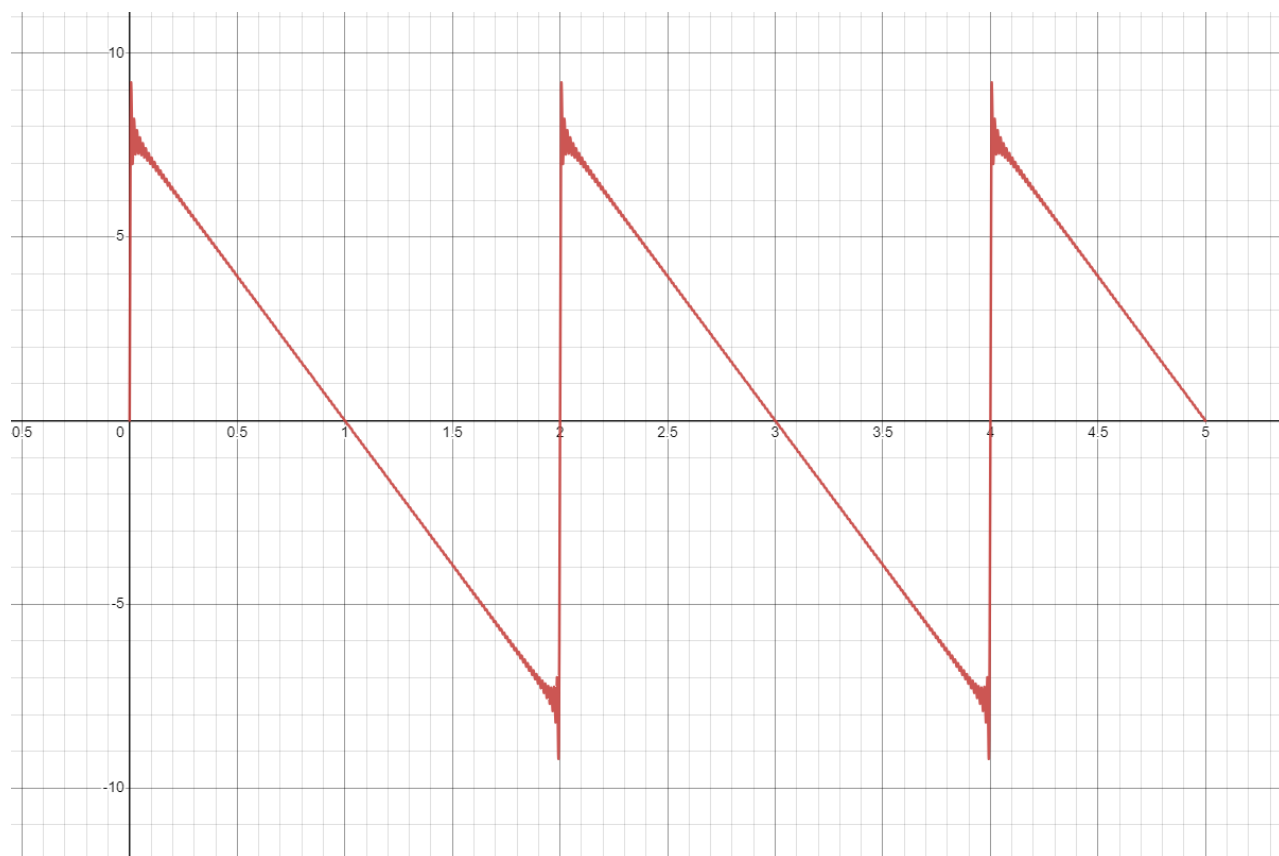
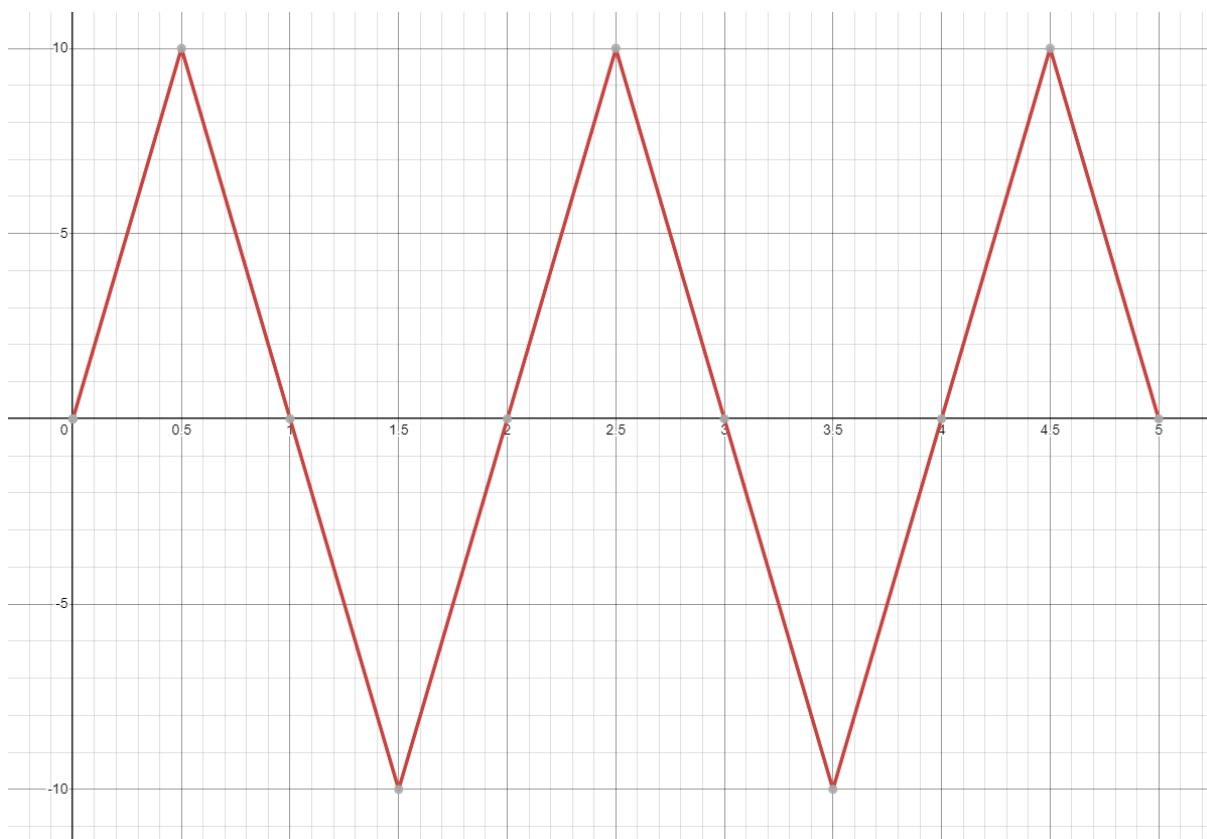


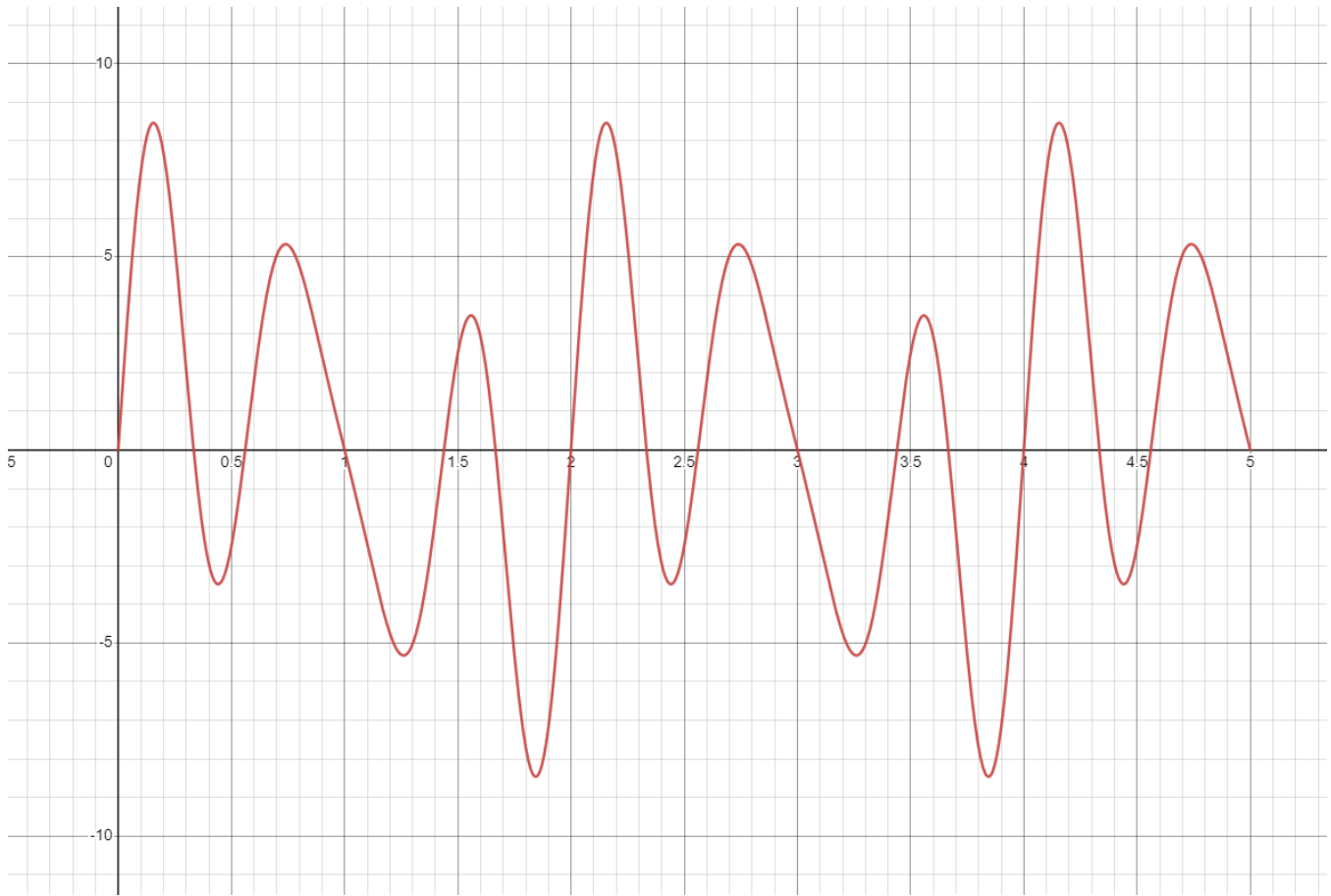
The output is now given by:



- 4) (CO1, 2.5 x 4 = 10) Now find the output waveform (V_{out}) for the following input waveforms (V_{in}). The comparator circuit is the same circuit used in case 2 examples:

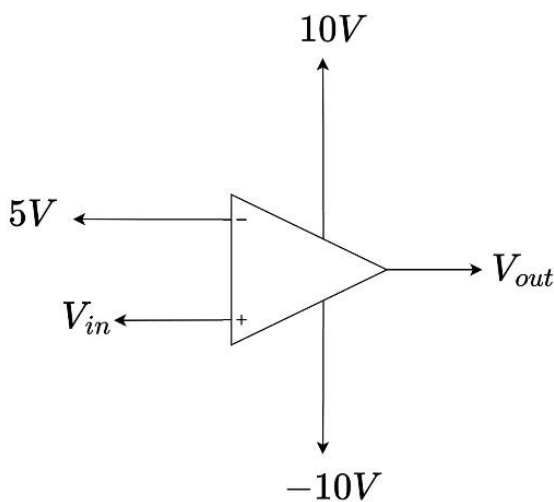






Case 3: $V_1 = V_{ref}$ DC (setting inverting input as a fixed DC value):

The op amp comparator would look something like:



Here, V_{ref} is a fixed DC voltage that we set in any one of our inputs (in this case, the inverting input) which is used as a baseline for comparison of our input voltage V_{in} . We essentially compare our input voltage to this reference voltage V_{ref} to see when its bigger or smaller than this reference. Here, V_{ref} set to be equal to 5V. So that, at all times, $V_1 = 5V$ DC and $V_2 = V_{in}$ that is our input waveform. If we write down the mathematical expressions for our comparator circuit, we get:

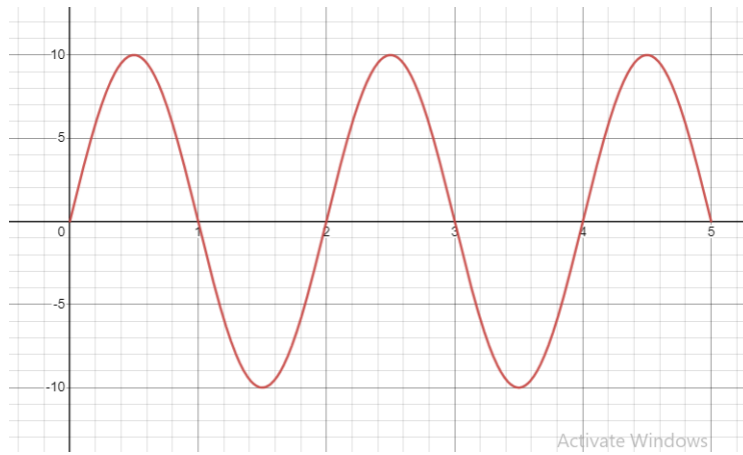
$$V_2 > V_1: V_{out} = 10V \text{ and } V_2 < V_1: V_{out} = -10V$$

From our circuit, we can now write:

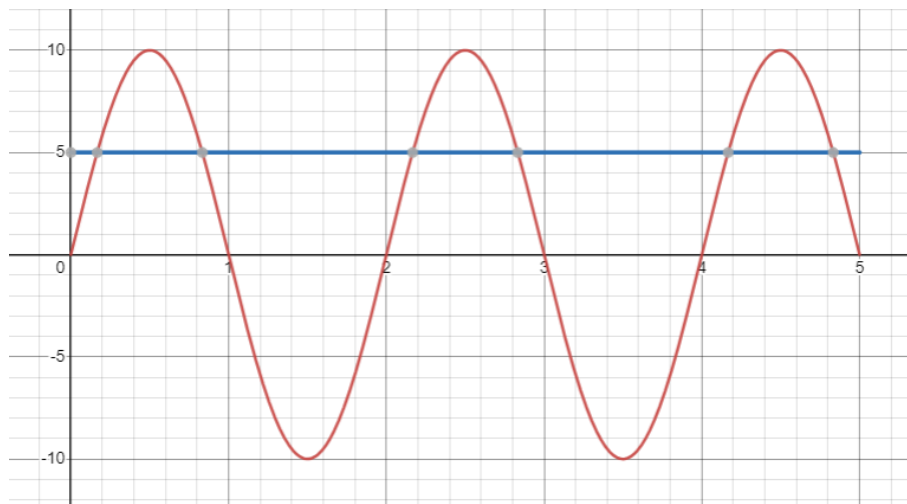
$$V_{in} > 5V: V_{out} = 10V \text{ and } V_{in} < 5V: V_{out} = -10V$$

Which means that the input is now compared to the fixed 5V DC Value. Whenever it goes bigger than 5V, output is 10V and whenever it remains smaller than 5V, output is -10V. Let us take a

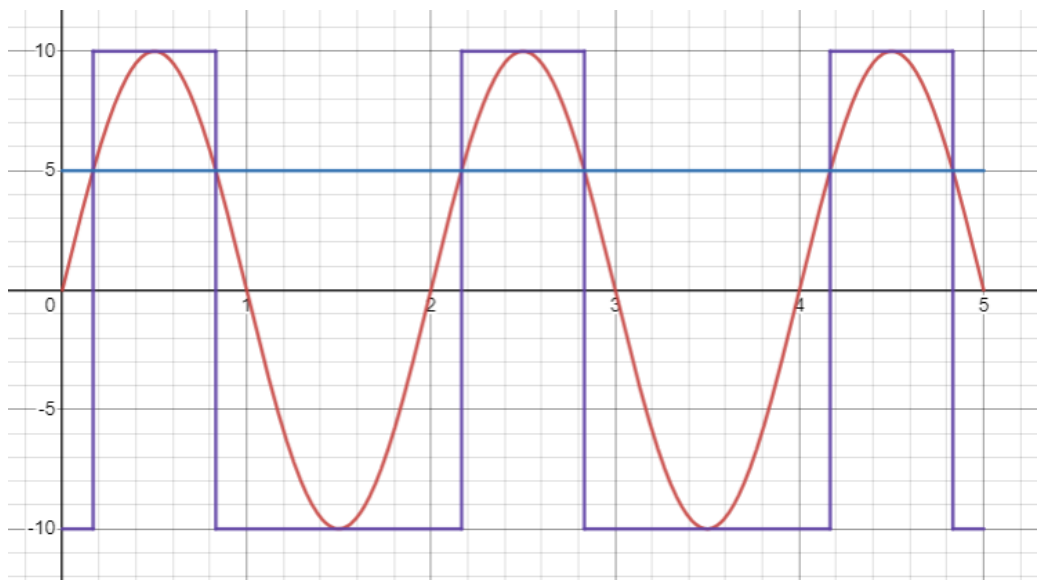
sinusoidal waveform of amplitude 10V as an example:



It's hard to tell when this waveform is bigger or smaller than 5V, so we draw the 5V DC as a waveform on top of this input waveform. Thus our graph becomes:

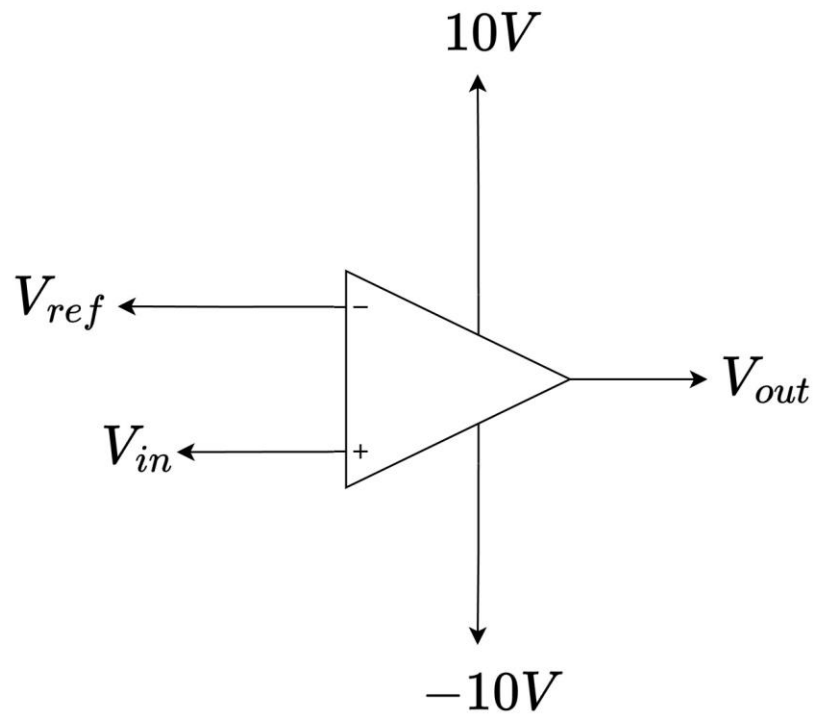


Now, we can compare when our input waveform (red graph) is bigger or smaller than 5V DC (blue graph). The output graph should come out to be something like this:

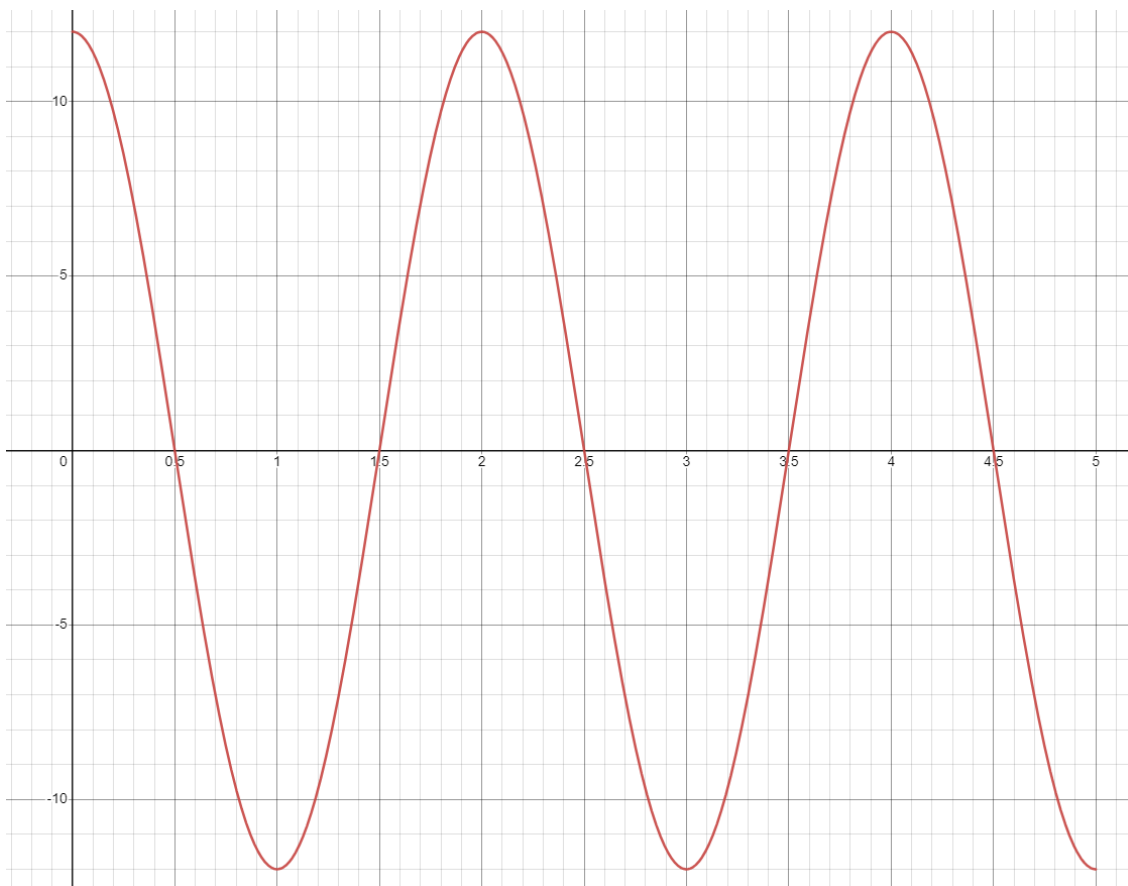


Yet again, notice how whenever the input waveform, shown in red, is greater than 5V, shown in blue, the output, shown in purple immediately goes to 10V. And whenever it is less than 5V, the output is immediately reaching negative saturation, i.e., going to -10V.

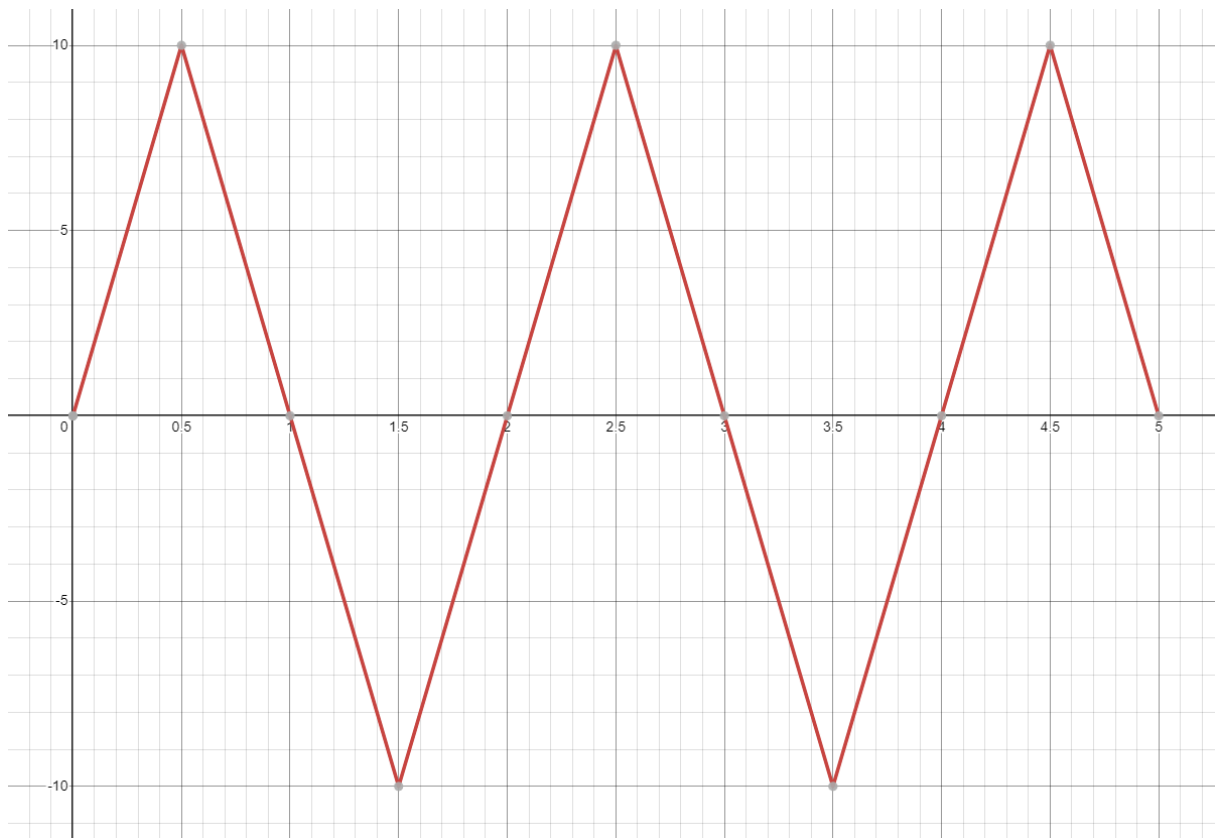
- 5) (CO1, 3 x 3 = 9) Now find the output waveform (V_{out}) for the following input waveforms (V_{in}). The V_{ref} value is given before the graph. Draw the V_{ref} DC waveform as well as shown in the examples and then draw the output waveform. The op amp circuit is as follows:



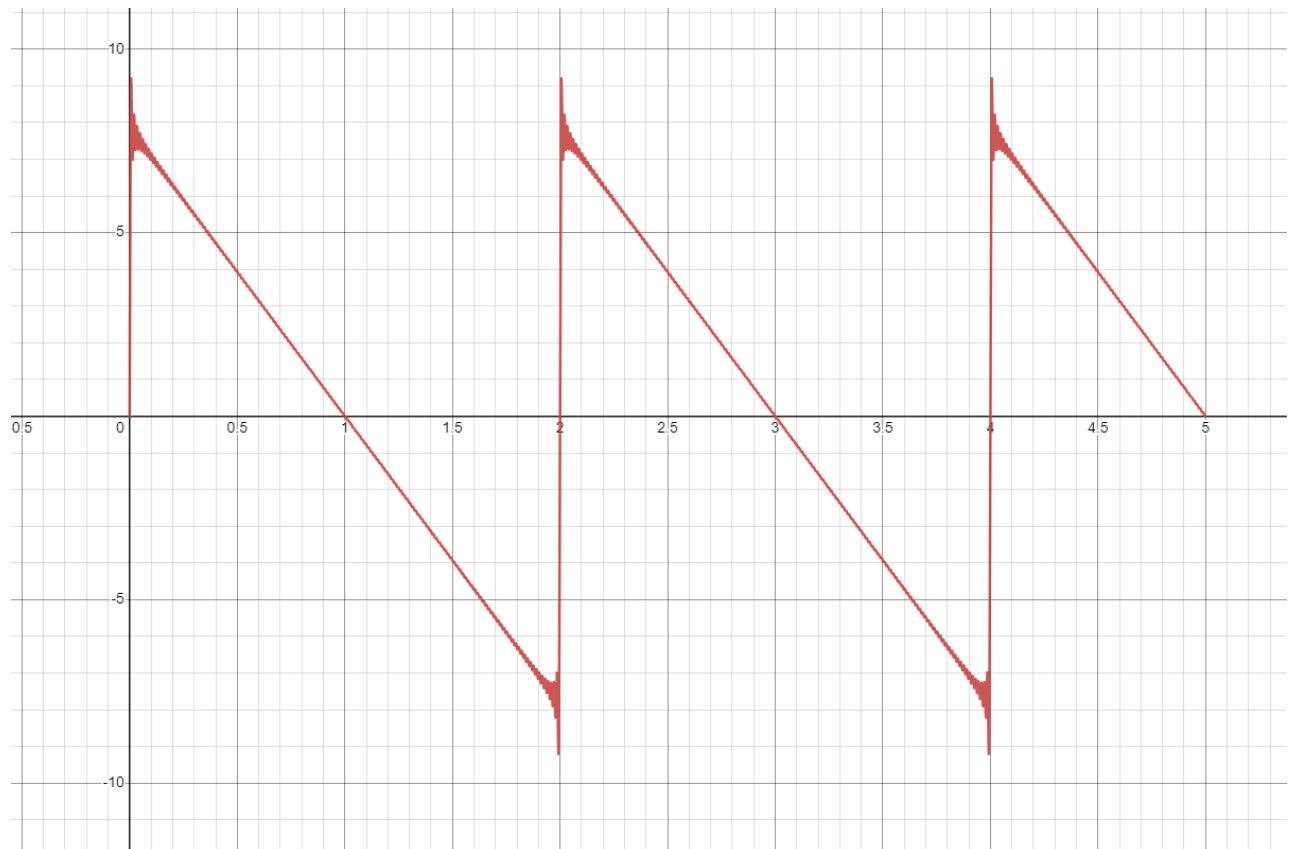
i) $V_{ref} = 2V$



ii) $V_{\text{ref}} = -5\text{V}$



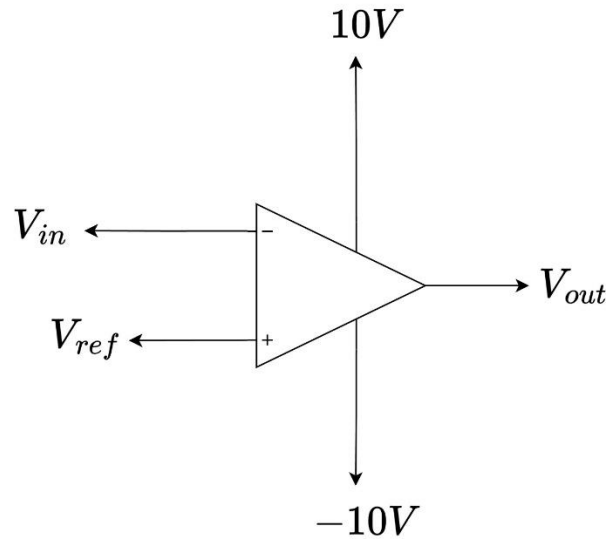
iii) $V_{\text{ref}} = 3\text{V}$



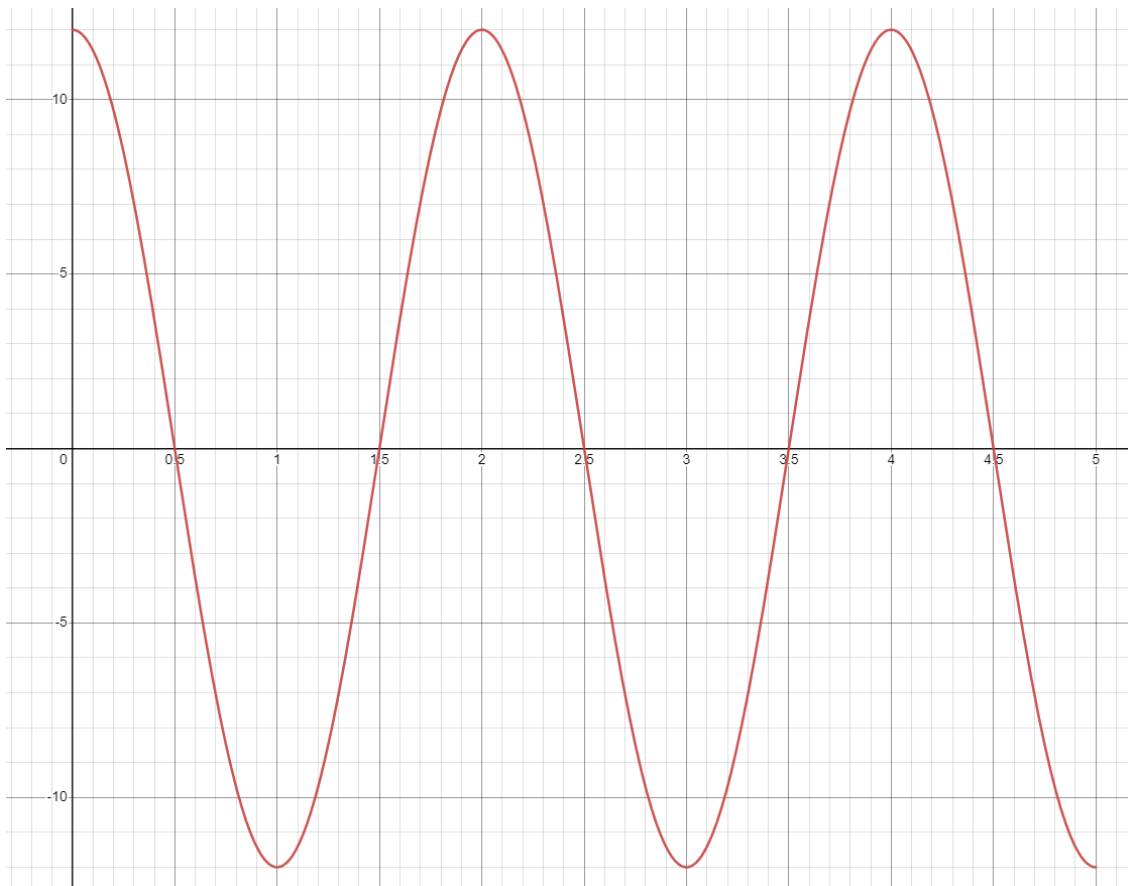
Case 4: $V_2 = V_{ref}$ DC (setting inverting input as a fixed DC value):

Exactly similar to case 3. Students should try to do this one themselves as it follows exactly the same analysis techniques as the previous cases.

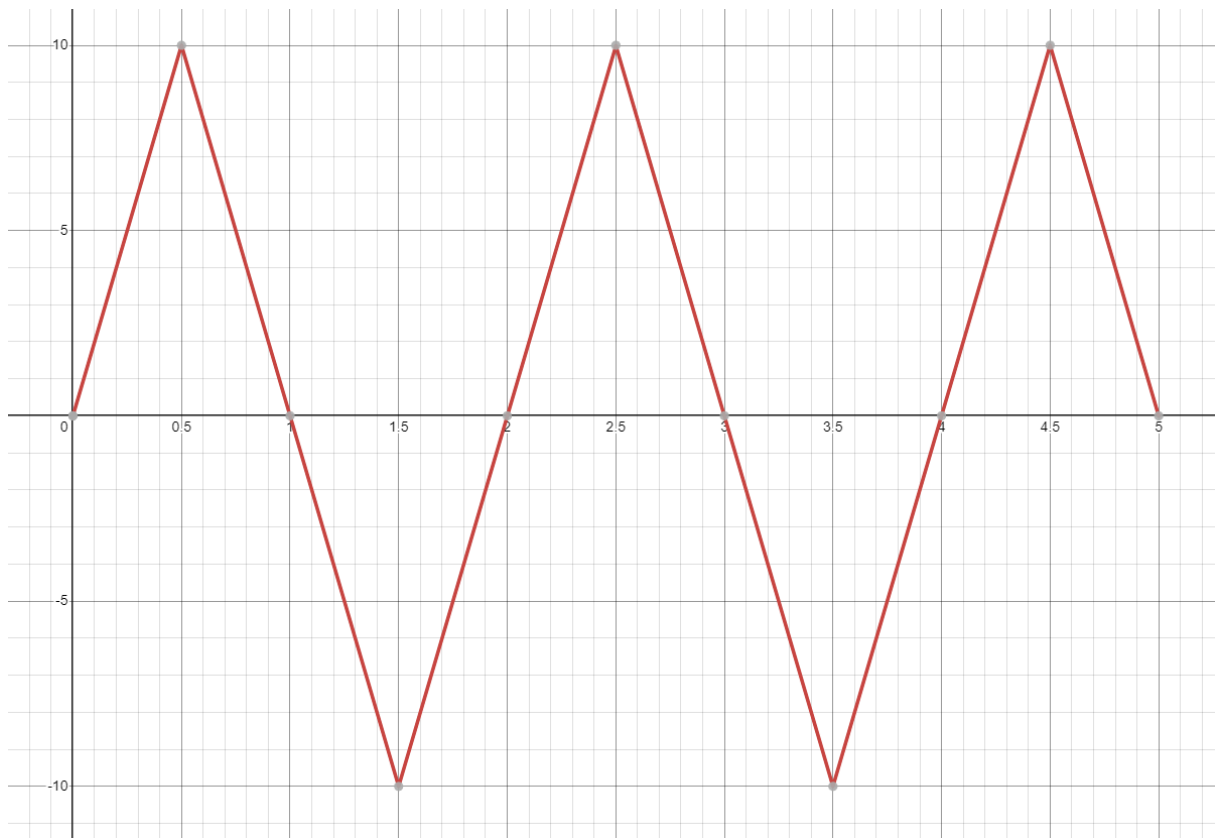
- 6) (CO1, 3 x 3 = 9) Find the output waveform (V_{out}) for the following input waveforms (V_{in}). The V_{ref} value is given before the graph. Draw the V_{ref} DC waveform as well as shown in the examples and then draw the output waveform. The op amp circuit is as follows:



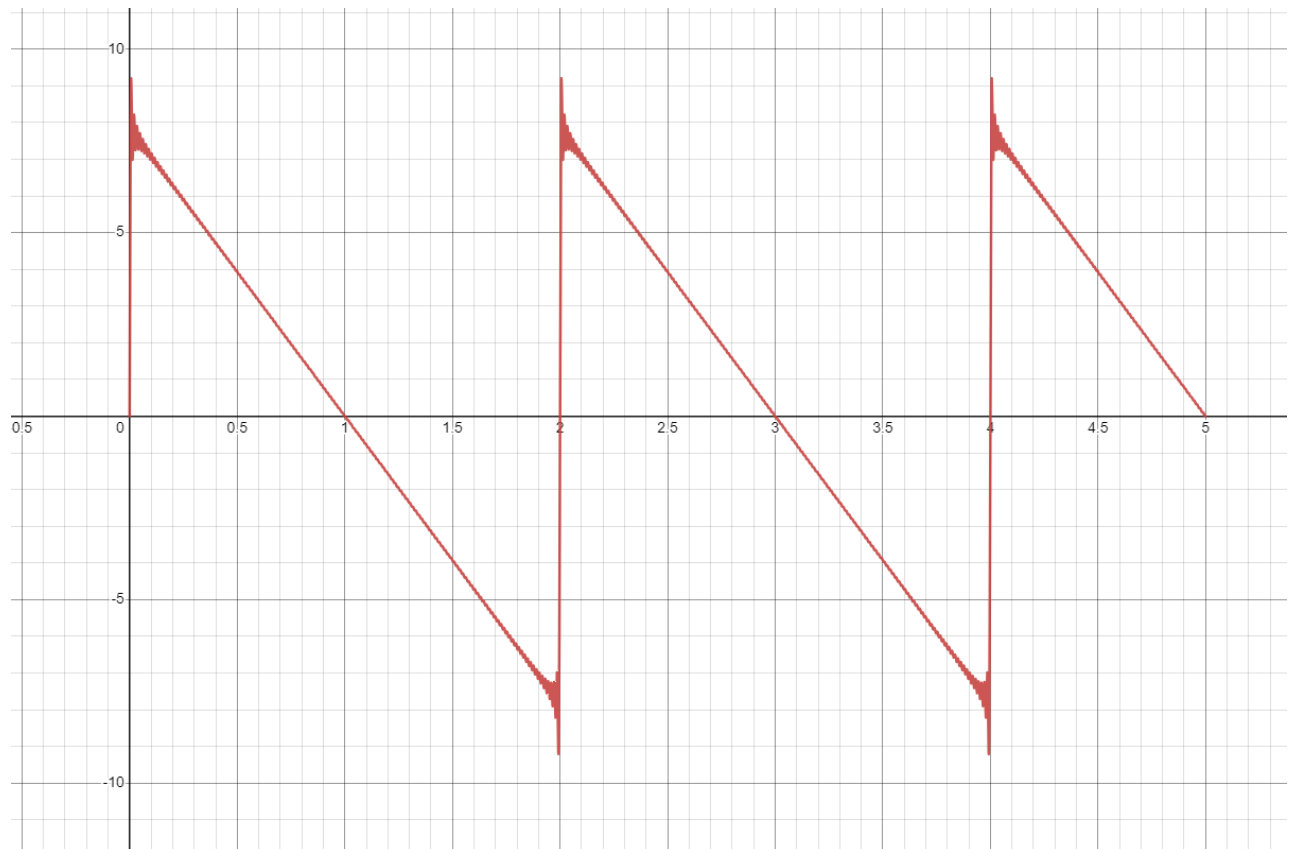
i) $V_{ref} = 2V$



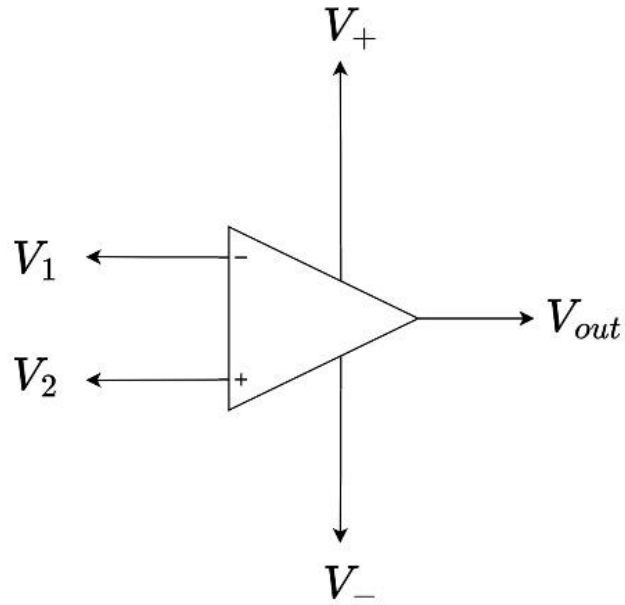
ii) $V_{\text{ref}} = -5\text{V}$



iii) $V_{\text{ref}} = 3\text{V}$



Case 5: An intricate case where both V_1 and V_2 can vary!



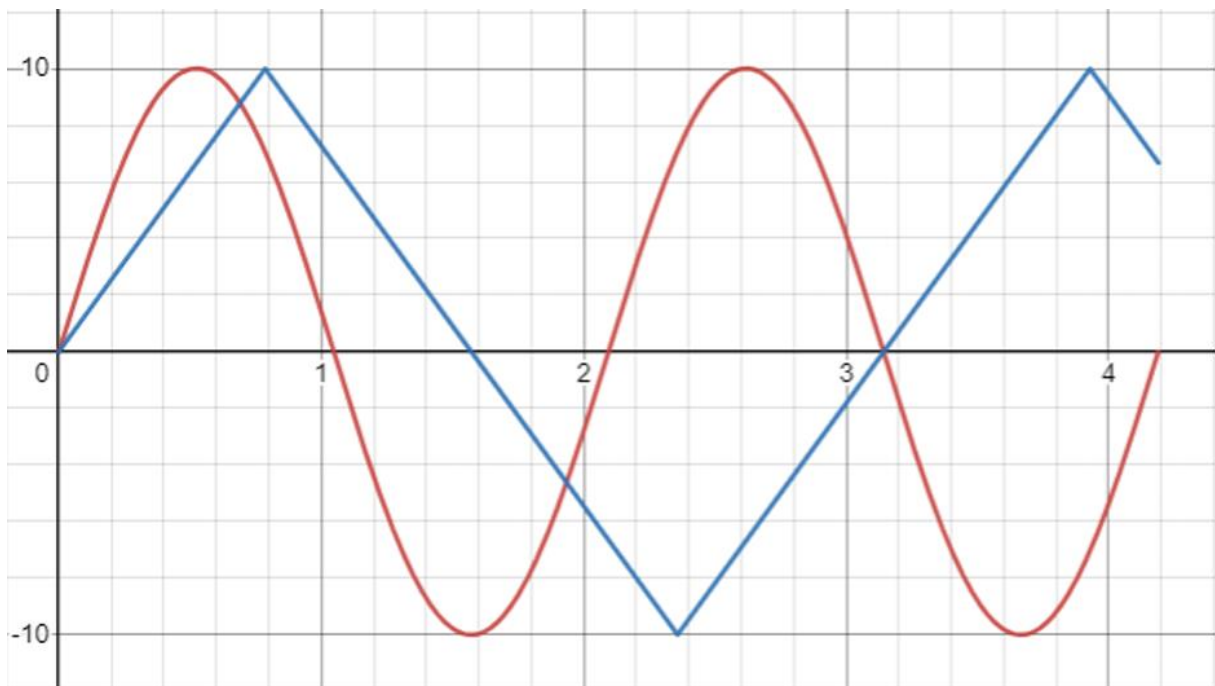
The op amp circuit can now have both V_1 and V_2 as varying waveforms. The solution is still simple. We first remember the two equations of the op amp comparator:

If $V_2 > V_1$ then, $V_{out} = V_+$

If $V_2 < V_1$ then, $V_{out} = V_-$

And we simply observe in the graph when V_2 is bigger than V_1 . For that duration, the output is positive saturation voltage. And during the periods when V_2 is smaller than V_1 , the output is the negative saturation voltage.

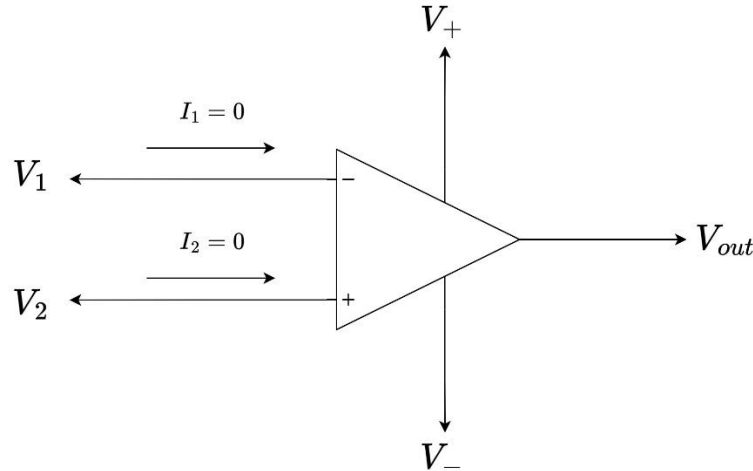
- 7) **(CO1, 7.5)** Try to draw the output waveform V_{out} for the following graph. V_1 is given by the blue waveform and V_2 is given by the red waveform:



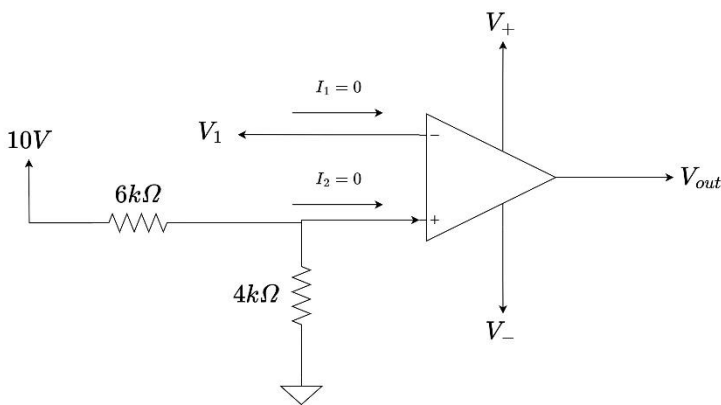
Op-Amp Closed Loop Configuration:

Before moving any further, we note that the operational amplifier draws very little current in its input terminals, almost close to the range of nanoamperes. Therefore, for all practical purposes, we approximate the current drawn by the op amp input terminals as zero.

$$\text{Therefore, } I_1 = I_2 = 0$$



This is a very useful property of the op amp, as it allows us to connect any node of any arbitrary linear network to any of the inputs and use the node voltage as input voltage to the op amp. An example is given below:



Here, a simple voltage divider network is connected to the op amp terminal V_2 . Since the input terminal does not draw any current, we can independently analyze this input network and find the node voltage of the node that is connected to the input terminal.

And we can find V_2 with voltage division law:

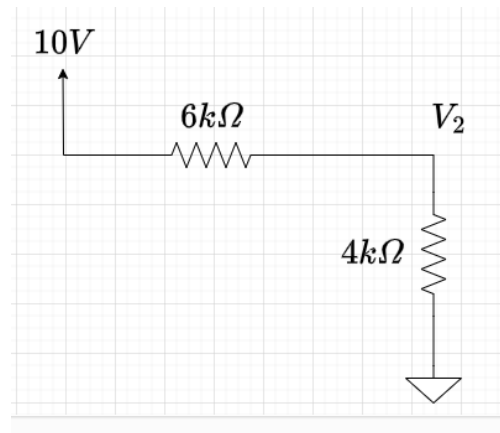
$$V_2 = (4k / (6k + 4k)) * 10V = 4V$$

Thus, the op amp now has input voltage $V_2 = 4V$.

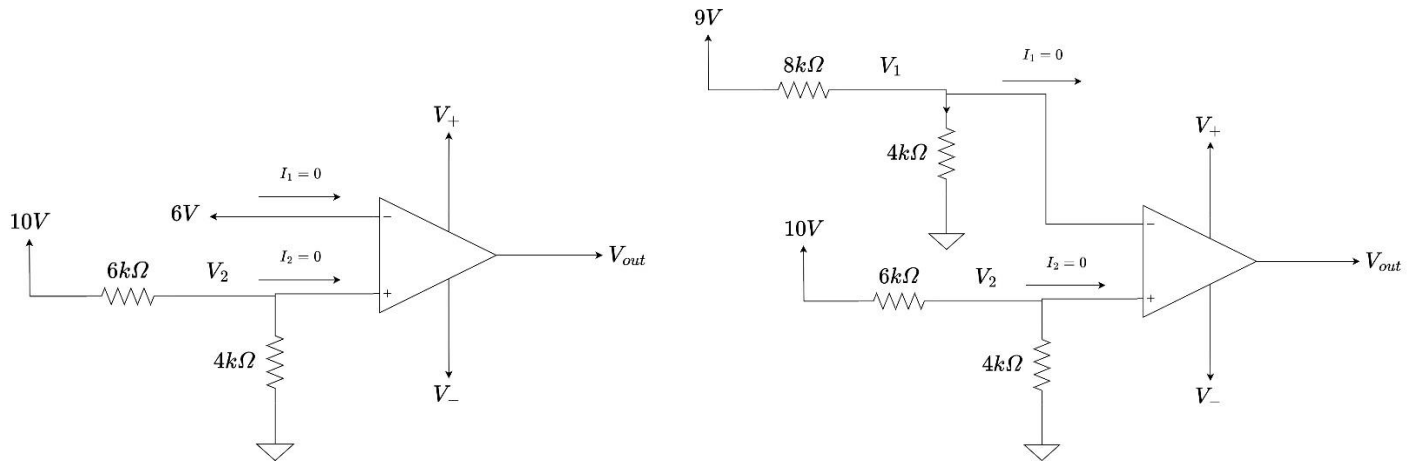
Then the op amp expressions turn out to be:

$$\text{If } 4V > V_1 \text{ then } V_{out} = V_+$$

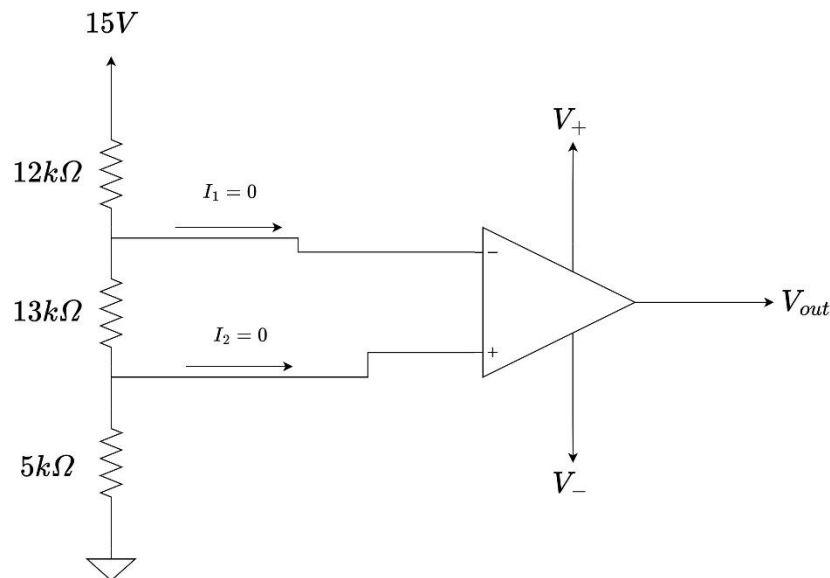
$$\text{If } 4V < V_1 \text{ then } V_{out} = V_-$$



8) (CO2, 2.5 x 2 = 5) Find the output of the following op amp circuits:



A more tricky example:



This example will be worked through. In order to solve this problem, we find the node voltages of the nodes connected to the op amps. By inspection, we see that it is a series circuit. With total current:

$$I = 15 / (12k + 13k + 5k) = 0.5 \text{ mA}$$

Then the voltage at V_1 is $(12k \times 0.5 \text{ mA})$ less than 15V:

$$= 15 - (12k \times 0.5 \text{ mA}) = 15 - 6 = 9V.$$

Thus, $V_1 = 9V$.

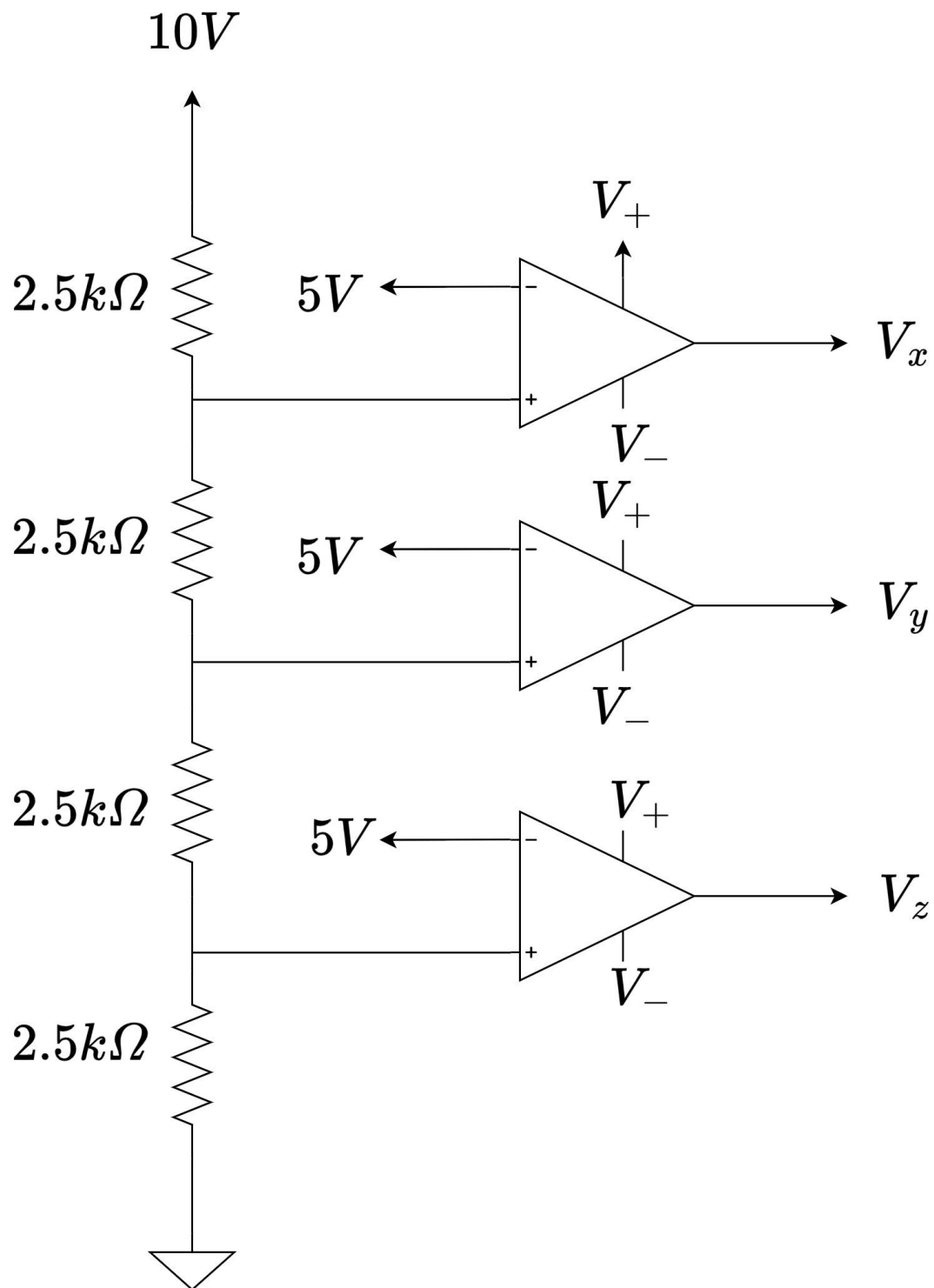
Whereas, voltage at V_2 is can be found by considering it is 15V minus two voltage drops at 12k and 13k so: $V_2 = 15V - (12k+13k) \times 0.5mA = 2.5 \text{ V}$.

So this op amp has $V_1 > V_2$: it's output, $V_{out} = V_-$.

The basic idea is to:

- 1) Isolate the input network connected to op amp input terminals in the comparator circuit and solve the node voltages
- 2) Find which nodes are connected to the input terminals and then find the op amp outputs.

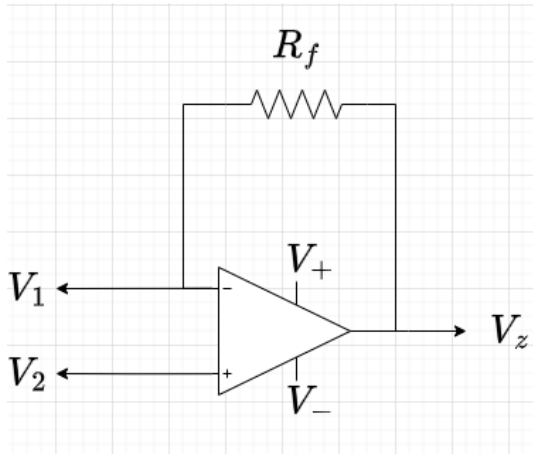
9) (CO2, 8) Solve the following op amp comparator circuit and calculate V_x , V_y and V_z with the ideas formulated:



Op Amp Closed Loop Configuration:

So far, we've seen that the operational amplifier can be configured to act as a comparator between two input voltages. But it can actually be used as an incredibly simple amplifier that amplifies any voltage signal/waveform. What makes it even more powerful is how the amplification can be adjusted with simple changes in resistance values.

In the closed loop configuration, the output is connected to the inverting input via a feedback resistor, R_f . The op amp looks something like this:



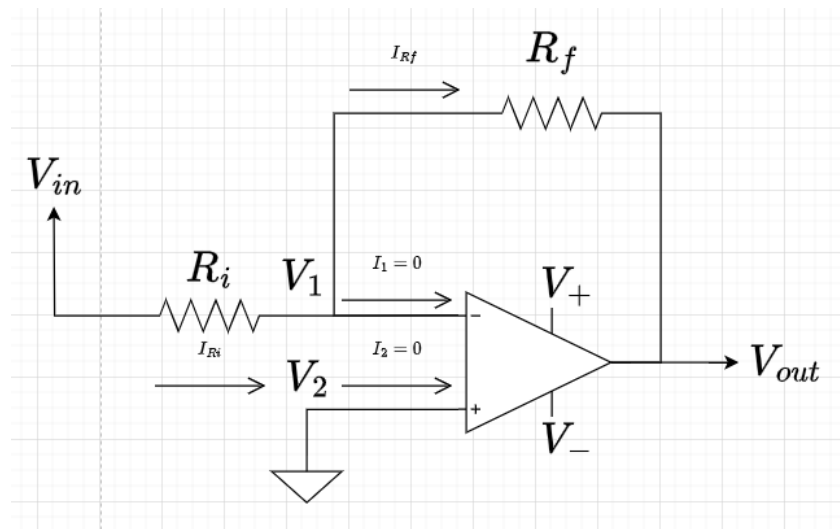
With this kind of configuration, if the inputs of the op amp are not forced to be any specific input voltage, the op amp tries to make both its input voltages to be equal to each other.

Thus, in most closed loop circuits, we have:

$$V_1 = V_2$$

We will use this assumption in solving all of our closed loop configurations of op amp with feedback resistors connected to input V_1 and output V_{out} .

a) Inverting Amplifier Configuration:



Here, by applying an input voltage and connecting to input terminal V_1 with an input resistance, R_i , the op amp can be allowed to force V_1 and V_2 to be equal. Since $V_2 = 0$, V_1 is also forced to be at ground potential so that,

$V_1 = V_2 = 0$. This grounding of V_1 without connecting directly to ground is called virtual ground.

Now if we perform KCL at node V_1 , we have:

$$(V_1 - V_{in}) / R_i + (V_1 - V_{out}) / R_f = 0$$

Considering $V_1 = V_2 = 0$ and solving for V_{out} , we get:

$$V_{out} = -(R_f / R_i) * V_{in}$$

This configuration now gives an amplified version of the input, the amplification of which depends on the ratio of the input and output resistances.

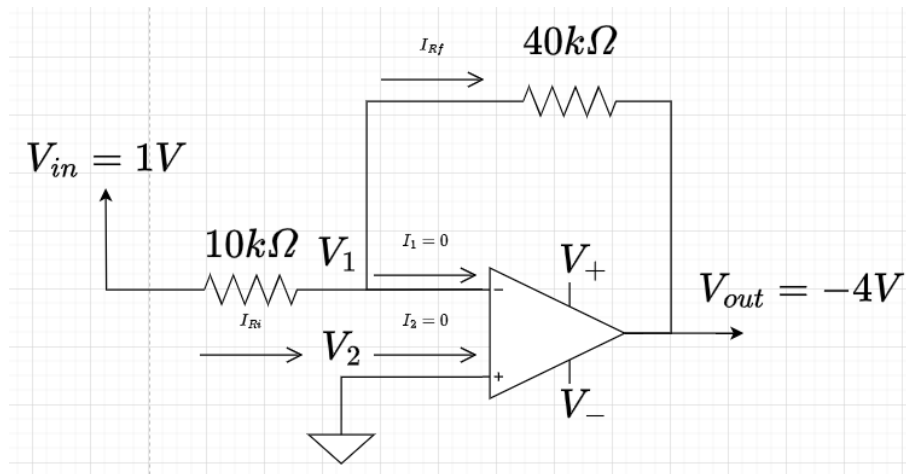
Here is an example:

Say, we have $R_f = 40k$ and $R_i = 10k$. Then the output voltage for this configuration will be:

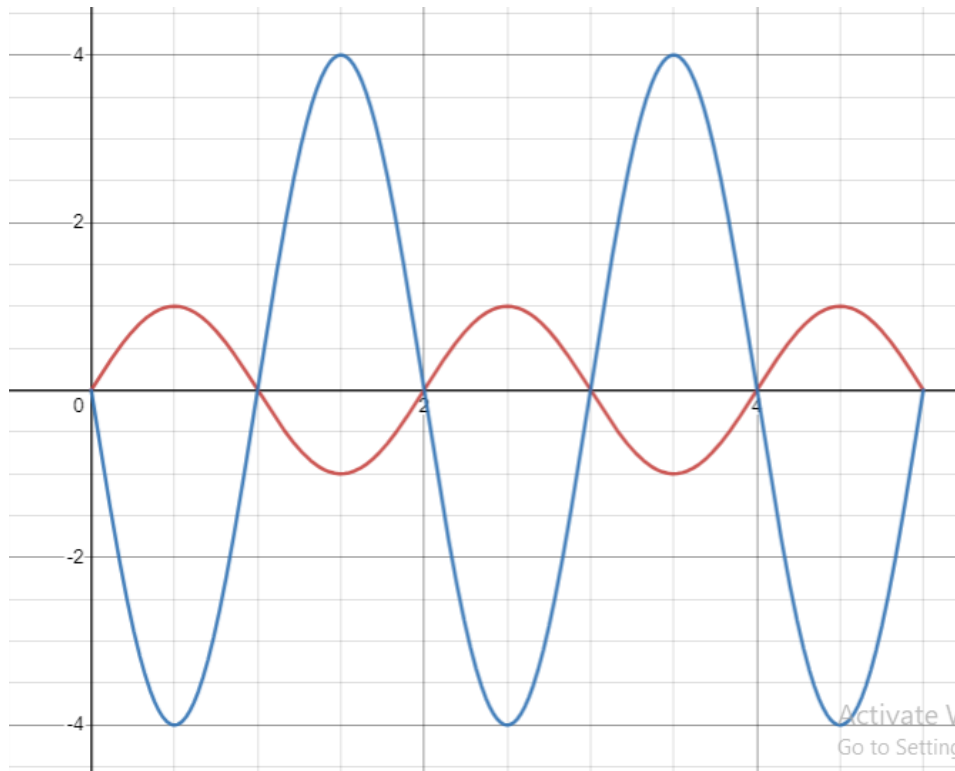
$$V_{out} = -(40k / 10k) * V_{in} = -4V_{in}$$

Thus, we have a 4 times bigger input voltage. Thus, if $V_{in} = 1V$, then $V_{out} = -4V$. However, the output is an inversion of the input as well.

However, we must remember that the op amp output is limited to the range: $V_- < V_{out} < V_+$. So, if the input were to be 5V and saturation voltages were at -15V and +15V, then output would not be $-4 * (5V) = -20V$ but rather it would be -15V.



But this is the output of a simple DC voltage. What about if we give an input waveform. Then we'll get 4 times the amplitude of that waveform. And of course, inverted as well. An example is given below:

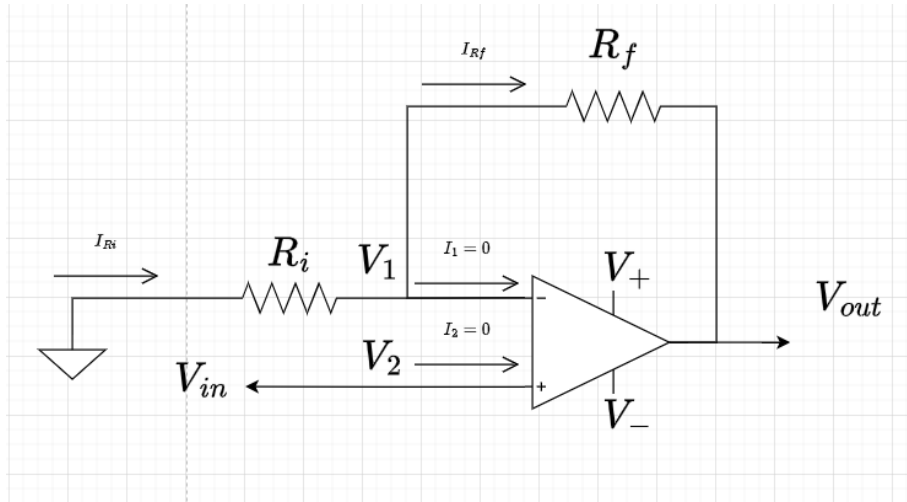


Here, red waveform => input voltage, V_{in}

Blue waveform => output voltage, V_{out}

b) Non-Inverting Amplifier Configuration:

This configuration is extremely similar to the previous inverting configuration, only difference is that the ground connection and the input voltage, V_{in} are swapped. Thus the diagram becomes:



In this configuration, $V_1 = V_2 = V_{in}$. Now, if we perform KCL at node V_1 (or, alternatively consider currents I_{Ri} and I_{Rf} using ohm's law and write them as equal), we can solve for V_{out} and obtain:

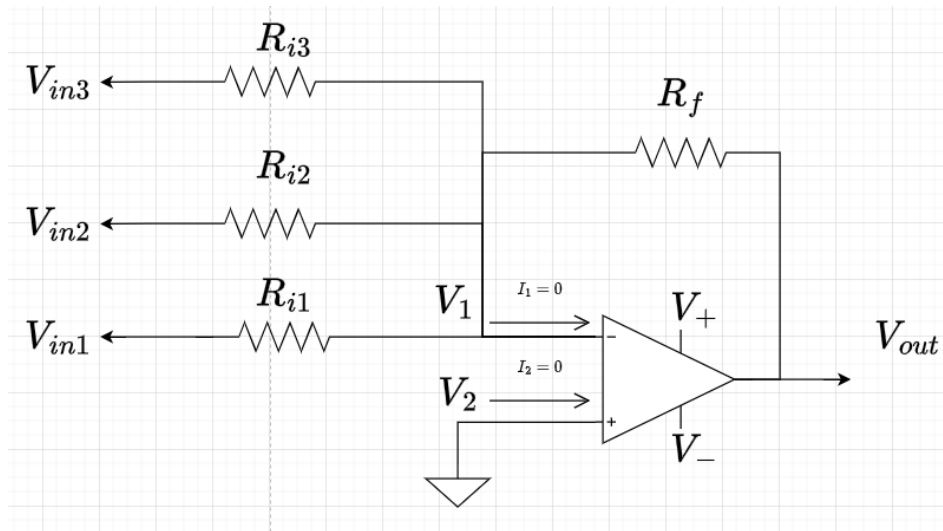
$$V_{out} = (1 + R_f/R_i) * V_{in}$$

Example: $R_f = 40k$, $R_i = 10k$, then V_{out} becomes:

$$V_{out} = (1 + 40k/10k)V_{in} = (1 + 4)V_{in} = 5V_{in}$$

c) Inverting Summer Configuration:

The inverting summer is as shown below:



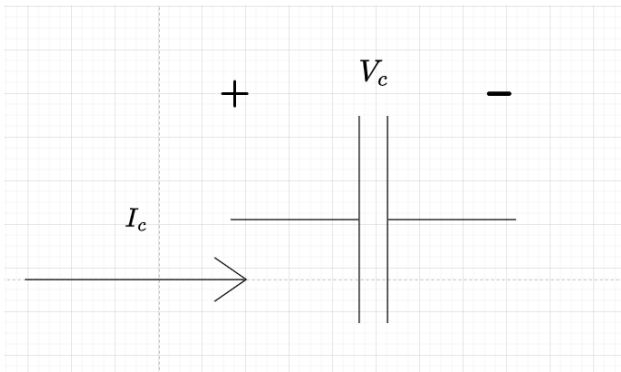
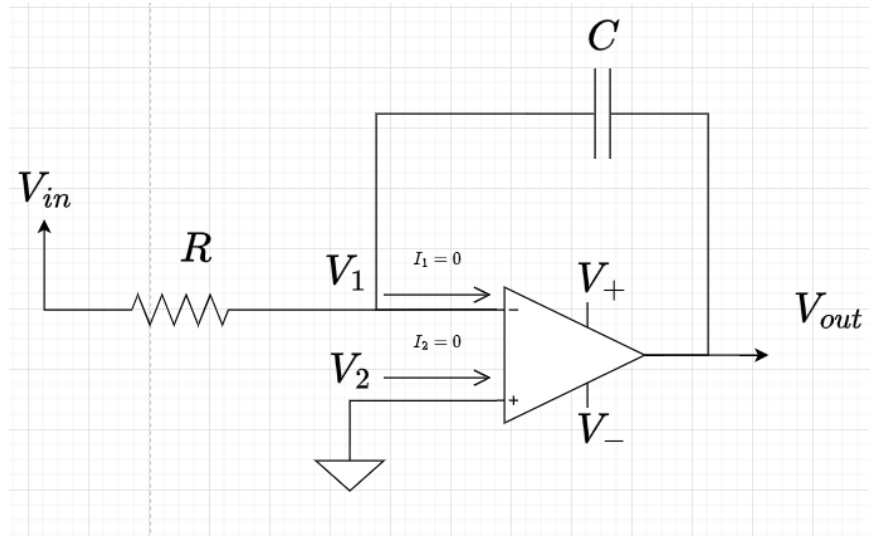
Try to find the output voltage expression. It will be as follows:

$$V_{out} = -[(R_f/R_{i1}) * V_{in1} + (R_f/R_{i2}) * V_{in2} + (R_f/R_{i3}) * V_{in3}]$$

This configuration can sum any number of inputs with a respective gain value obtained from the feedback resistor and input resistor ratio.

d) Integrator:

The feedback resistor is replaced with a capacitor:



We know, the current through capacitor is related to its voltage by:

$$I_c = C[d(V_c)/dt]$$

Since current through R and C is the same [$I_1 = 0$], thus we have:

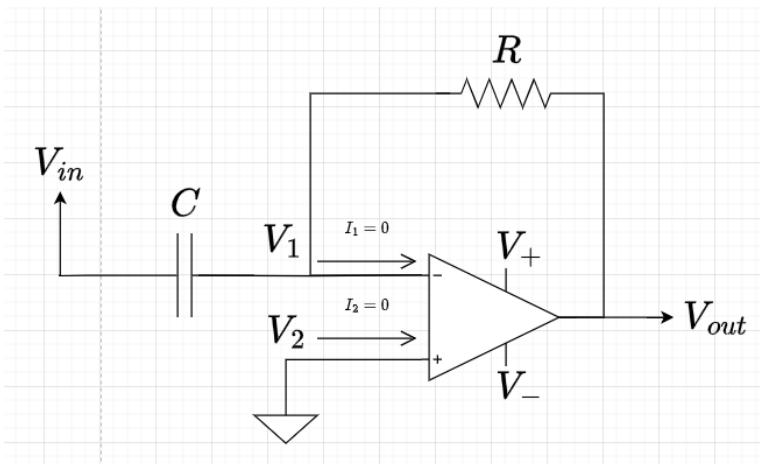
$$(V_{in} - 0)/R = I_c = C[d(0 - V_{out})/dt]$$

$$\text{or, } d(-V_{out})/dt = V_{in}/RC$$

$$\text{or, } V_{out} = -[\int V_{in} dt] / RC$$

e) Differentiator:

The differentiator is given as:



The differentiator output is given as:

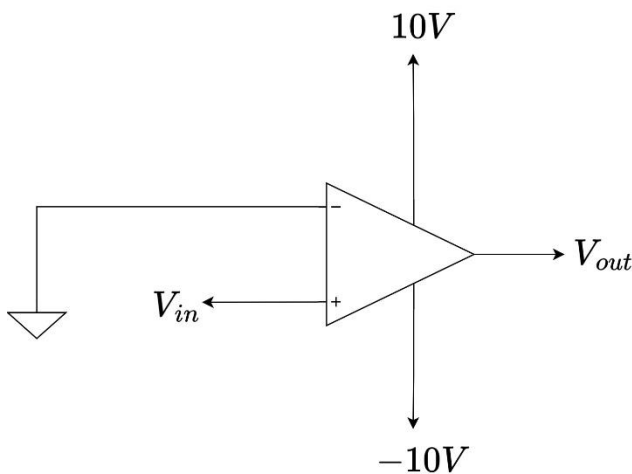
$$V_{out} = -RC[d(V_{in})/dt]$$

- 10) (CO2, 2 x 5 = 10)** Derive the output expressions of the five op amp closed loop configurations and write them down in your copy
- 11) (CO2, 3 x 2 = 6)** Find the output expressions and draw a rough waveform in your copy for the following given specifications:
- Inverting amplifier: $R_f = 50k$, $R_i = 10k, 50k, 100k$. $V_{in} = 2V$ peak to peak sine wave
 - Noninverting amplifier: $R_f = 50k$, $R_i = 10k, 50k, 100k$. $V_{in} = 2V$ peak to peak sine wave
 - Summing amplifier: $R_f = 50k$, $\{R_{i1}, R_{i2}\} = \{10k, 50k\}, \{50k, 50k\}, \{100k, 50k\}$. $V_{in1} = 2V$ peak to peak sine wave and $V_{in2} = 1V$ DC.

Voltage Transfer Characteristics:

Voltage transfer characteristics curve is an input voltage vs output voltage graph where the input voltage remains in the x-axis and the output voltage in the y-axis. The output voltage V_{out} is plotted as a function of the input voltage V_{in} .

- a) Comparator with $V_1 = 0$:

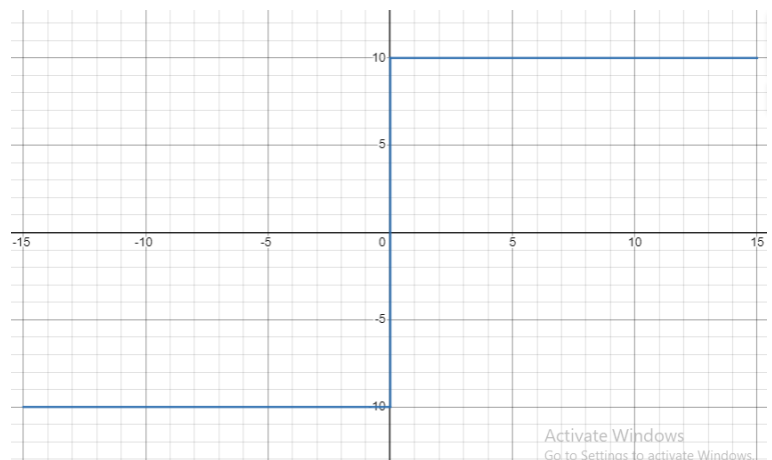


We know, the output is given as:

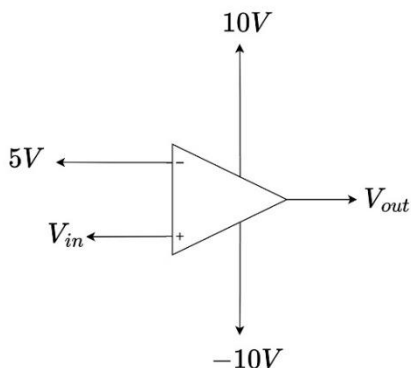
$$\text{If } V_{in} > 0 \text{ then: } V_{out} = 10V$$

$$\text{If } V_{in} < 0 \text{ then: } V_{out} = -10V$$

We can plot the V_{out} as a piecewise function of V_{in} . The graph should look like the one given below:



- b) Comparator with $V_1 = 5V$:



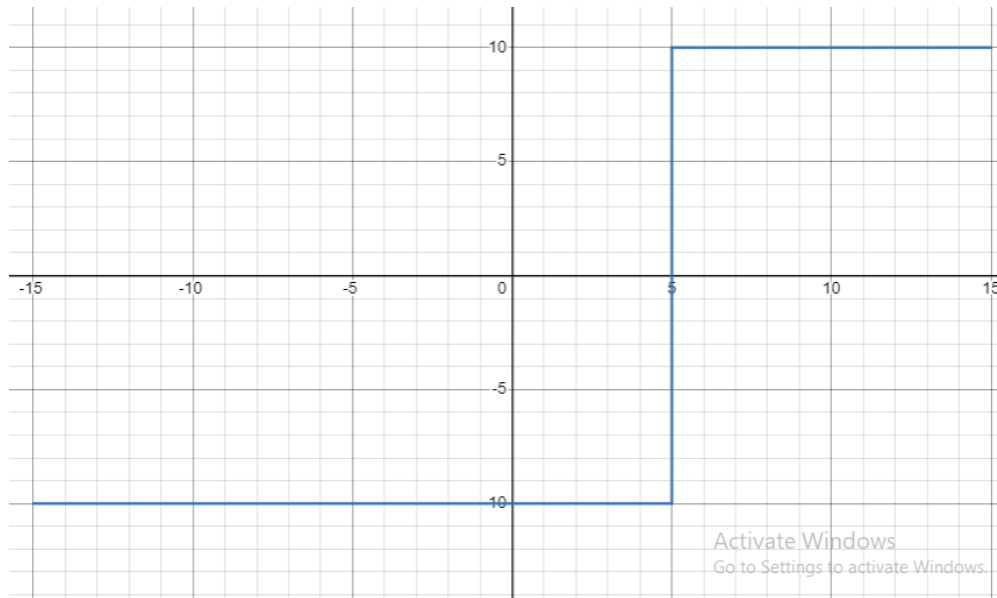
For this circuit, we write the output expressions and inspect:

$$\text{If } V_{in} > 5V \text{ then: } V_{out} = 10V$$

$$\text{If } V_{in} < 5V \text{ then: } V_{out} = -10V$$

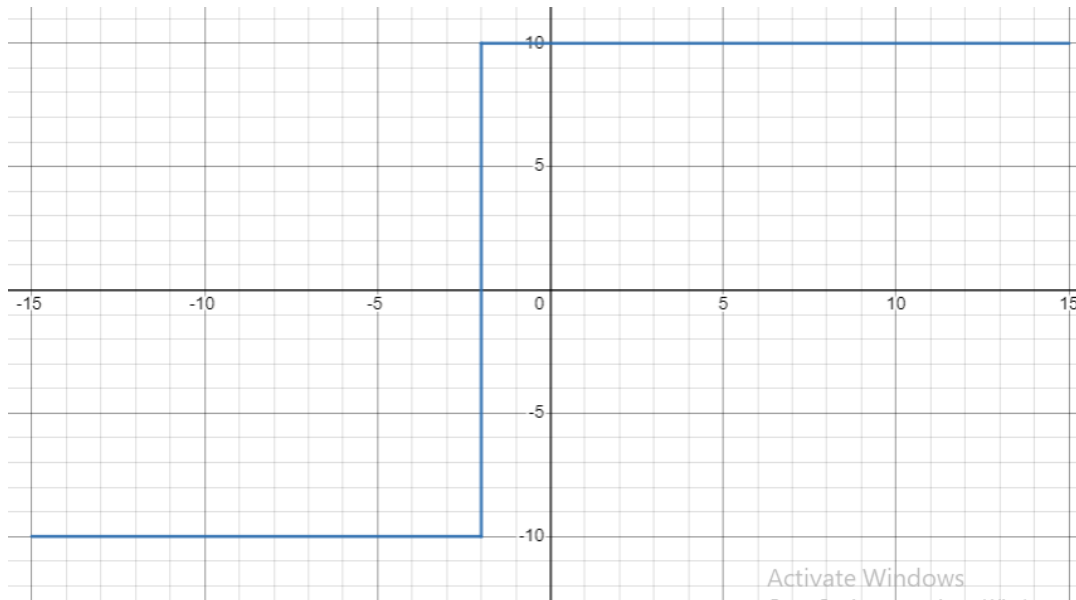
And the voltage transfer characteristics curve is given by:

[Notice how the output voltage now transitions at 5V instead of 0V. That makes sense as per our output expressions where it abruptly changes at 5V]

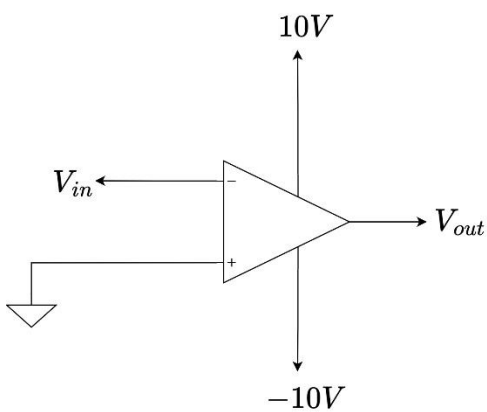


c) Comparator with $V_1 = -2V$:

The VTC with $V_1 = -2V$ would look something like this:



d) Comparator with $V_2 = 0V$:

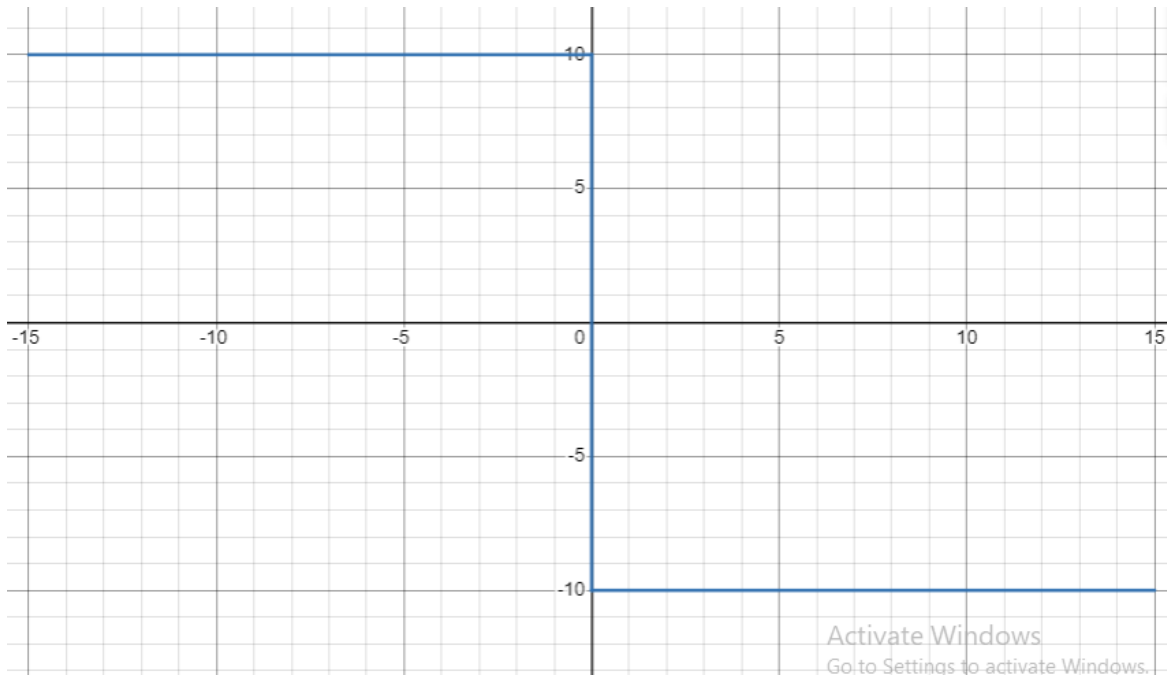


$0V > V_{in}$: $V_{out} = 10V$ and $0V < V_{in}$: $V_{out} = -10V$

Which can be further rewritten as:

$V_{in} < 0V$: $V_{out} = 10V$ and $V_{in} > 0V$: $V_{out} = -10V$

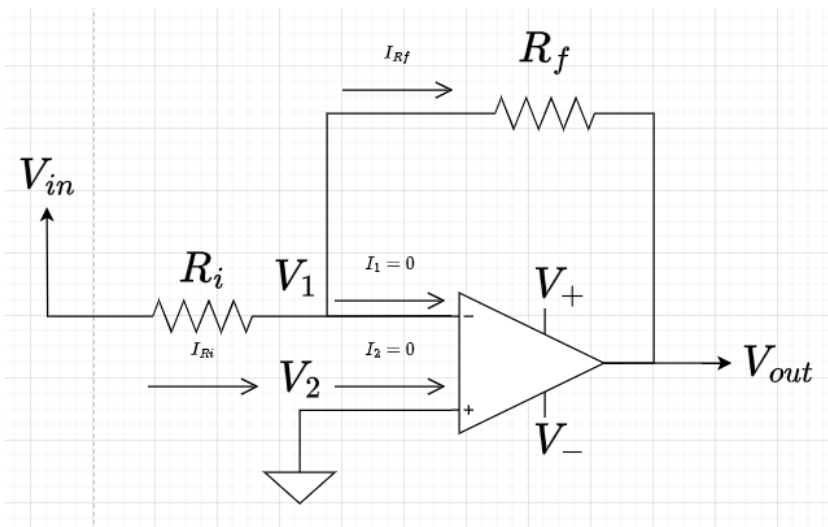
The voltage transfer characteristics can now be drawn as:



12) (CO2, 0.5 x 14 = 7) Draw the VTC curve of the following comparator circuits:

- Comparator with $V_1 = -3V, -2V, -1V, 0V, 1V, 2V, 3V$
- Comparator with $V_2 = -3V, -2V, -1V, 0V, 1V, 2V, 3V$

e) Now we consider the inverting configuration:



The output is given by:

$$V_{out} = -(R_f/R_i)V_{in}$$

If we take $R_f/R_i = 4$, we get:

$$V_{out} = -4V_{in}$$

If we treat V_{in} as x and V_{out} as y [since the input voltage is the x -axis and output voltage is the y -axis], we get the line equation:

$$y = -4x$$

This is simply a straight line, and if we draw this straight line, our VTC is done! However we

must remember that output is bounded by V_+ and V_- . Thus:

$$V_- < V_{out} < V_+$$

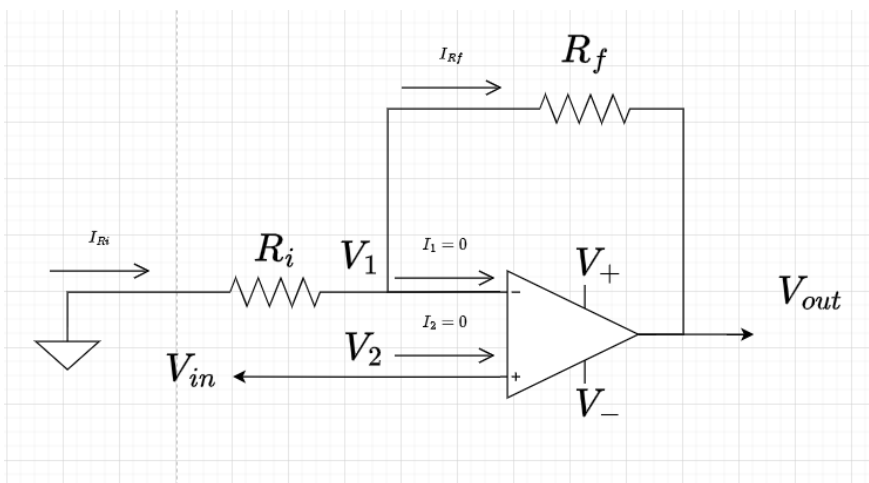
For now, let us say that these lower and upper bounds are given by $-15V$ and $+15V$ respectively. Then the graph is a straight line bounded within -15 and 15 in the y -axis.

The VTC is then given as:



The straight line is between -15 and 15 and within that limit, it is a straight line with slope -4.

f) Now let us see the non-inverting configuration:



The output voltage expression is given by:

$$V_{out} = (1 + R_f/R_i)V_{in}$$

Where, $V_- < V_{out} < V_+$

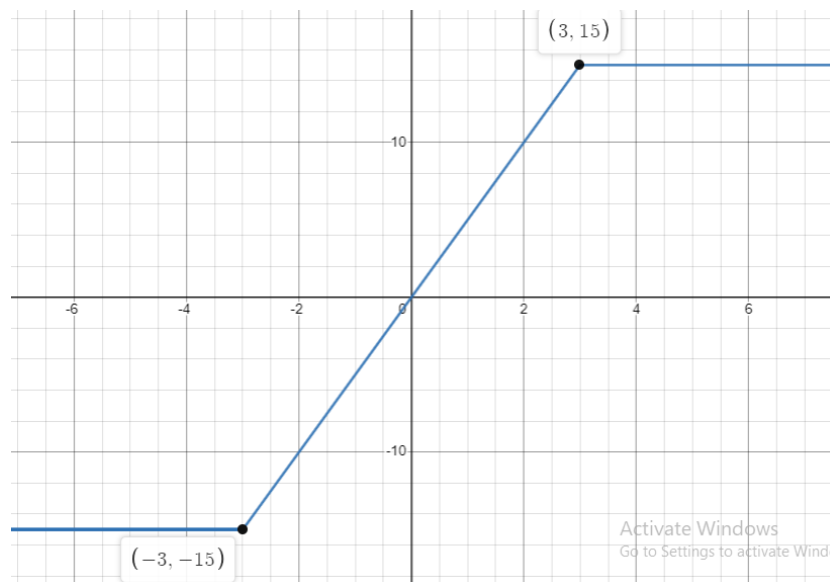
Considering $R_f/R_i = 4$, and $V_- = -15V$, $V_+ = 15V$, we have:

$$V_{out} = 5V_{in}$$

Which if rewritten with x and y:

$$y = 5x$$

The VTC comes out to be:



- 13) (CO2, 0.5 x 14 = 7)** Try to now draw the VTC of the following configurations. Consider $V_- = -15V$, $V_+ = 15V$ (A rough sketch with key values pointed out in graph will be more than enough):
- Inverting amplifier: $R_f = 50k$, $R_i = 10k, 50k, 100k$.
 - Noninverting amplifier: $R_f = 50k$, $R_i = 10k, 50k, 100k$.

[Prepared by TMY: Tanvir Muntakim Tonoy]

[Submission Guidelines: Must be submitted in the google form given in the discord channel. Deadline is two weeks from uploading of problem set]

